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Role of Carbonaceous Supports and Potassium Promoter on Higher Alcohols Synthesis over Copper−Iron Catalysts

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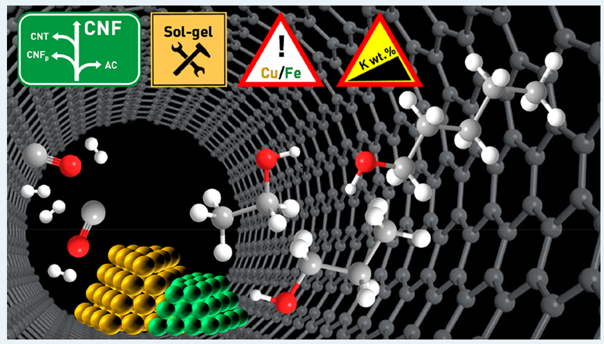
Supporting Information



ABSTRACT: The identification of an effective copper−iron catalyst for the direct conversion of synthesis gas into higher alcohols is hindered by the low solubility limit of Cu in Fe and the limited understanding of structural and electronic descriptors in such multicomponent systems. Here, commercial carbonaceous carriers are shown to produce an efficient material only if they enable control of the size and location of metal species through confinement in adequately sized channels, with conical carbon nanofibers being more adequate than carbon nanotubes. Application of a sol−gel route was preferred to other deposition methods to avoid excessive Cu aggregation, associated with enhanced CO_2 formation. A bulk Cu/Fe ratio of 2 permitted one to balance the different tendencies of Cu and Fe toward agglomeration, i.e., to form numerous Cu particles of moderate size and hinder the dispersion of the Fe phase and in turn the Fischer−Tropsch activity. Promotion by tiny amounts of potassium was instrumental to further increase the size of the Fe particles and enhance their proximity to Cu. These structural features were associated with a more facile Cu-mediated reduction of the Fe phase and a more pronounced hydrogen activation ability, based on thermal characterization under reaction conditions, and maximized the higher alcohols selectivity (47%) and the olefins fraction among hydrocarbons (50%). An in-depth kinetic analysis over the top performer provided guidelines to optimize temperature, pressure, H_2/CO ratio, and residence time, leading to a space time yield of 0.53 $g_{HA}\ g_{cat}^{-1}\ h^{-1}$. This value is almost twice as high as that of the state-of-the-art bimodal silica-supported CuFe system and could be maintained for 100 h on stream.

KEYWORDS: *higher alcohols synthesis, syngas, copper−iron catalysts, nanostructured carbon supports, potassium promotion*

1. INTRODUCTION

Higher alcohols (HA), defined herein as alcohols possessing two or more carbon atoms, have a wide range of applications in the (fine) chemical industry and the energy sector.[1] Their production relies on sugar fermentation (C_2OH) and alkene hydration ($C_{3+}OH$). These technologies suffer from high dilution, which implies a high energy input for product separation, and low single-pass conversion, respectively. The direct chemocatalytic transformation of synthesis gas (syngas, CO + H_2), derived from coal, natural gas, or lignocellulosic biomass feedstocks, to HA is regarded as an alternative synthesis pathway holding potential to bridge the gap between sustainability and market demand. Still, the design of a sufficiently effective catalyst comprises one of the key challenges in the development of an industrially amenable technology.[2] Besides for Rh-based catalysts, forming Rh^0 and Rh^+ species,[3] distinct metals have been applied to cover the C−C coupling and CO hydrogenation functions required,[4] i.e., CuCo,[5] CuFe,[6] and CoMo.[7,8] HA with a backbone of three or more C atoms is more easily attained over these bimetallic materials, which are also more economical.

Systems based on Cu and Fe, individually applied to methanol synthesis (MS)[9] and Fischer−Tropsch synthesis (FTS),[10] respectively, have been extensively investigated.[11−16] The alloying of the two metals is thermodynamically hindered (only 2.7 atomic % of Cu in Fe even at 1123 K).[17] Thus, they typically remain as separated phases. Upon activation in H_2 and exposure to syngas, both metals are reduced into their metallic forms, with Fe further transforming to different types of iron carbides (Fe_xC) to a substantial extent. On the basis of experimental evidence and density function theory (DFT) calculations, Lu et al.[6] suggested that the reaction occurs

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selectively at the Cu^0–χFe_5C_2 interface. While it remains controversial if other types of Fe carbides in contact with Cu may lead to superior performance, fostering the interaction between the two metals is the main target.

Different synthetic strategies have been followed to enhance the contact between Cu and Fe in bulk systems.[14,18] For instance, macroporous materials derived from a colloidal crystal template method attained a CO conversion of 58% and a HA selectivity of 26% at a gas hourly space velocity (*GHSV*) of 2000 h^{-1}.[19] Metal promoters have also been used to provide structural effects.[20–22] Lin et al.[21] claimed that Mn could enhance the dispersion of both Cu and Fe, while Zn could stabilize Fe against sintering through the formation of $ZnFe_2O_4$. When Ding et al.[20] added K to the CuFeZnMn system, they observed migration of Fe from the bulk to the surface of the material. Still, these solids reached a maximum HA selectivity of 31%. With respect to supported CuFe catalysts, a layered double hydroxide precursor was adopted to enhance the dispersion and intimacy of Cu and Fe over MgO,[13,23] which led to a CO conversion comparable to the best bulk system (57%) and a higher HA selectivity (37%) at the same *GHSV*. Alternatively, the particle size of the metals and their proximity were tuned by tailoring the porosity of the carrier, a bimodal silica.[11,24–26] The best K-promoted CuFe catalyst supported on such materials (33 wt % CuFe) achieved a CO conversion of 56% and a HA selectivity of 46% at 6000 h^{-1}.[26] Its superiority to other reported systems was attributed to the adequate size of the pores, which offered surface to boost metals dispersion but avoided the formation of too small precursor particles, the reduction of which would be hindered by strong metal–support interactions. Yet, the metals particle size was only assessed in the fresh materials, and the role of the K promoter was not clarified.

To shed further light onto the effect of the dimensions and intimacy of Cu and Fe particles, carbonaceous materials stand as a versatile platform of supports to derive solids featuring distinct structural and morphological properties. Indeed, activated carbon (AC) and platelet-type carbon nanofibers (CNF_p) possess high surface areas promoting a fine distribution of the metals, while carbon nanotubes (CNT) and conical carbon nanofibers (CNF) have a hollow-core structure which offers the possibility to limit the growth of metals particles due to confinement effects.[8,27–30] To date, only N-doped CNT has been adopted as a carbon-based carrier for Cu and Fe.[31] This catalyst displayed a moderate performance (HA selectivity = 19%, CO conversion = 21%), but the work emphasized the beneficial effect of basicity by N doping to shift the alcohols distribution from methanol (MeOH) to HA.

Herein, we compare CuFe catalysts supported on various carbonaceous materials (i.e., CNT, CNF, CNF_p, and AC) with a low total metal content (5 wt %), identifying CNF as the best carrier. Using this support, we vary the Cu/Fe ratio, add K as a promoter in different amounts, and apply distinct synthesis method to produce solids featuring different properties. In-depth characterization by electron microscopy, X-ray photoelectron spectroscopy, and high-pressure temperature-programmed desorption of H_2 shed light onto the role of support and the promoter on the morphology, size, and location of the active metal species. To gather insights into the reaction kinetics, the best catalyst is evaluated at different temperatures (493–593 K), pressures (3–7 MPa), weight hourly space velocities (*WHSV*, 4000–32 000 $cm^3\ g_{cat}^{-1}\ h^{-1}$), and molar H_2/CO feed ratios (0.5–4). Finally, the catalyst stability is also examined monitoring its performance during a 100 h run.

2. EXPERIMENTAL SECTION

2.1. Catalyst Preparation. A bulk CuFe catalyst with nominal molar Cu/Fe = 2 was prepared by coprecipitation using Na_2CO_3 as the precipitating agent. $Cu(NO_3)_2 \cdot 3H_2O$ (5.283 g, Aldrich Fine Chemicals, 98–103%) and $Fe(NO_3)_3 \cdot 9H_2O$ (4.406 g, Aldrich Fine Chemicals, >98%) were dissolved simultaneously in deionized water (65.6 cm^3) yielding a solution with a total metal concentration of 0.5 M. The mixture was magnetically stirred at 343 K while adding a 0.5 M Na_2CO_3 solution (81–83 cm^3, 2 $cm^3\ min^{-1}$) to reach a pH of 8. The slurry obtained was aged for 3 h. Then the solid was filtered, washed with water (3 dm^3), dried at 343 K, calcined at 723 K, and reduced in a 10 vol % H_2/He flow of 20 $cm^3\ min^{-1}$ for 4 h at 673 K (ramp rate = 3 K min^{-1}).

Supported CuFe catalysts with nominal molar Cu/Fe ratios of 1, 2, 3, or 5, and a CuFe loading of 5 wt % were prepared by a sol–gel method (SG). K was added as a promoter to the materials with Cu/Fe = 2 in a nominal molar K/(Cu + Fe) ratio of 0.005, 0.01, or 0.1. CNF (Aldrich-Fine Chemicals, Fe content < 100 ppm), CNF_p (ABCR, >98%), CNT (ABCR, 95%, outer diameter = 20–30 nm), and AC (ACROS Organics Co., Ltd.) were used as carriers as received. $Cu(NO_3)_2 \cdot 3H_2O$ (2.130–3.404 g) and $Fe(NO_3)_3 \cdot 9H_2O$ (1.138–3.562 g) were dissolved in deionized water (11.5 cm^3) under magnetic stirring. With the addition of 4–5 drops of NH_4OH (Acros Organics, 25 wt % aqueous solution), a dark green precipitate was formed. A citric acid (Sigma-Aldrich, ≥95%) solution (1.30–1.36 g in 3 cm^3 of water, corresponding to a molar citric acid/CuFe ratio of 0.4) was then added under magnetic stirring to redissolve the solid. When desired, an adequate amount of a K_2CO_3 solution (0.006–0.12 g in 1 cm^3 of water) was incorporated. The pH of the sol was adjusted to 3.5 adding a few drops of formic acid (Merck product, 98–100%) and NH_4OH. An aliquot of this final mixture was added to 2.0 g of support to achieve the desired metals loading. The slurry was magnetically stirred for 6 h at 338 K and dried in air at the same temperature overnight. The obtained solid was sieved to attain a 0.05–0.12 mm fraction, which was activated by reduction in a 10 vol % H_2/He flow of 20 $cm^3\ min^{-1}$ for 4 h at 673 K (3 K min^{-1}). These catalysts were coded combining information on the support (abbreviation), the molar Cu/Fe ratio, and the molar relative amount of K in the catalyst (if relevant), e.g., CNF-2-0.01. A portion of the CNF-2 catalyst was calcined in air for 3 h at 573 K (5 K min^{-1}) prior to the reduction step (denoted as CNF-2-C). The sol–gel method was applied to prepare additional catalysts supported on CNF and with nominal Cu/Fe = 2 through sequential addition of the metals, i.e., Fe/Cu/CNF-2 (Cu incorporated prior to Fe), K/CuFe/CNF-2-0.005 (K added after the simultaneous deposition of Cu and Fe), Fe/CuK/CNF-2-0.005 (Fe added after the simultaneous incorporation of Cu and K), Cu/FeK/CNF-2-0.005 (Cu added after the simultaneous incorporation of Fe and K), and CuFe/K/CNF-2-0.005 (Cu and Fe added simultaneously after K deposition). The amount of citric acid used during the individual deposition of Cu or Fe was adjusted to attain the same molar ratio as upon the one-pot deposition of the two metals. The solids were reduced after each incorporation of Cu and/or Fe applying the conditions detailed above.

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Supported catalysts with nominal molar Cu/Fe = 2 and CuFe loading of 5 wt % and CNF as the carrier were alternatively prepared by a wet deposition−reduction approach using $NaBH_4$ as the reducing agent (denoted as CR-CNF-2) and by deposition−precipitation using Na_2CO_3 as the precipitating agent (denoted as DP-CNF-2). Along the former route, $Cu(NO_3)_2 \cdot 3H_2O$ (0.2773 g) and $Fe(NO_3)_3 \cdot 9H_2O$ (0.2318 g) were dissolved in deionized water (172 cm^3) to attain a total metals concentration of 0.01 M. A 2-g amount of CNF was added to this aqueous solution, and the mixture was magnetically stirred at 298 K for 1 h. Afterward, a 0.5 M $NaBH_4$ solution was introduced (51 cm^3, 2 cm^3 min^{-1}), and the mixture was kept under stirring overnight at 298 K. The solid was recovered by filtration, washed with water (3 dm^3), and dried in vacuum overnight at 343 K. DP-CNF-2 was prepared by dissolving the same amounts of metals salts in 86 cm^3 of water, leading to a total metals concentration of 0.02 M. A 2 g amount of CNF was introduced to this solution, and the mixture was refluxed under stirring at 333 K for 1 h. Thereafter, a 0.04 M Na_2CO_3 solution was added (55 cm^3, 2 cm^3 min^{-1}) to the mixture until a pH of 8 was attained. The slurry was aged at the same temperature for 3 h. The solid was recovered by filtration, washed with water (3 dm^3), dried overnight in air at 333 K, calcined in air for 4 h at 723 K (5 K min^{-1}), and finally reduced under the aforementioned conditions.

2.2. Catalyst Characterization. The K, Cu, and Fe contents were determined by inductively coupled plasma optical emission spectrometry (ICP-OES) using a Horiba Ultra 2 instrument equipped with a photomultiplier tube detector. X-ray fluorescence spectroscopy (XRF) was performed using an Orbis Micro-EDXRF spectrometer equipped with a Rh source operated at 35 kV and 500 μA and a silicon drift detector to obtain the bulk Cu/Fe ratio in CP-2. N_2 sorption at 77 K was measured in a Micromeritics TriStar II instrument after degassing the samples at 573 K under vacuum for 3 h. The surface area of catalysts and supports was determined by applying the BET method. Powder X-ray diffraction (XRD) was conducted using a PANalytical X'Pert Pro-MPD diffractometer with Ni-filtered Cu Kα radiation (λ = 0.1541 nm), acquiring data in the 10−70° 2θ range with an angular step size of 0.033° and a counting time of 8 s per step. The size of metallic copper and iron crystallites was determined using the Scherrer equation. (High-pressure) temperature-programmed desorption of H_2 ((HP-)H_2-TPD) was carried out using a Micromeritics Autochem 2950 HP unit equipped with a thermal conductivity detector and coupled to a Pfeiffer Vacuum Omnistar GSD-320 quadrupole mass spectrometer. Reduced catalysts (0.250 g) were treated in a 10 vol % H_2/N_2 flow of 10 cm^3 min^{-1} at 0.5 MPa and 573 K for 3 h (3 K min^{-1}) to simulate the activation. A reference Fe/CNF sample was reduced at 773 K, due to the poor reducibility of Fe in the absence of Cu. After cooling down to the reaction temperature (543 K), the reducing gas was replaced by 60 vol % H_2/He (18.5 cm^3 min^{-1}) for 30 min, followed by He purging at a flow of 20 cm^3 min^{-1} for 1.5 h. H_2-TPD curves were acquired while heating up to 1073 K (5 K min^{-1}) in a the same He flow. For the high-pressure analysis, the pressure was increased to 5 MPa when admitting 60 vol % H_2/He after sample reduction and cooling. The adsorption and He purging times were equal to the ambient-pressure experiments. For temperature-programmed reduction with H_2 (H_2-TPR), CNF and catalysts in reduced and used forms (0.100 g) were dried in He flow of 20 cm^3 min^{-1} at 473 K for 1 h (3 K min^{-1}). After cooling down to room temperature in He, a 10 vol % H_2/Ar flow of 20 cm^3 min^{-1} was applied for 0.5 h. Methane formation and H_2 consumption were monitored using the mass signals (m/z) 15 (CH_3 fragment from CH_4) and 2, respectively, while heating up to 923 K (5 K min^{-1}). X-ray photoelectron spectroscopy (XPS) was conducted using a Physical Electronics (PHI) Quantum 2000 X-ray photoelectron spectrometer featuring monochromatic Al Kα radiation, generated from an electron beam operated at 15 kV and 32.3 W, and a hemispherical capacitor electron-energy analyzer, equipped with a channel plate and a position-sensitive detector. The energy scale of the instrument was calibrated using Au and Cu as reference samples. The catalysts were firmly pressed onto indium foil patches, which were then mounted onto a sample plate and introduced into the spectrometer. The analysis was conducted at 1 × 10^{-6} Pa, with an electron take-off angle of 45° and a pass energy of 46.95 eV. Charge compensation during the measurement was achieved using a low-energy electron source. Surface elemental concentrations were determined from the photoelectron spectra after Shirley background subtraction using the instrument specific sensitivity factors for calculation. The measurements were repeated after conducting sputtering with Ar^+ (1 kV potential) on an area of approximately 2 × 2 mm^2 for 0.5 min. High-resolution transmission electron microscopy (HRTEM) and scanning transmission electron microscopy coupled to energy-dispersive X-ray spectroscopy (STEM-EDX) were conducted in a FEI Talos F200A instrument equipped with a high-brightness field emission gun, a high-angle annular dark-field (HAADF) detector, and a large collection angle EDX detector, operated at 200 kV. Catalyst powders were dispersed on nickel grids coated with a continuous carbon film. Raman spectroscopy was carried out using a confocal Raman microscope (WITec CRM 200) equipped with a 532 nm diode laser and a 100× objective lens, operated in the backscattering mode with a power of 6 mW.

2.3. Catalyst Testing. The direct conversion of syngas to HA was carried out in a continuous-flow fixed-bed reactor setup, which is described in detail elsewhere.[8] A 0.5 g amount (sieve fraction = 0.05−0.12 mm) of undiluted catalyst was loaded into the reactor and purged with an Ar (Messer, purity 5.0) flow of 100 cm^3 min^{-1} for 0.5 h at ambient pressure. Under the same flow, the pressure was then increased to 5 MPa and a leak test was conducted. Thereafter, the catalyst was activated by flowing 10 vol % H_2/Ar at the same rate as for the inert gas previously used at 0.5 MPa and 573 K (3 K min^{-1}) for 3 h. The reaction was carried out by feeding a mixture of H_2 (Messer, purity 5.0), CO (Messer, purity 5.0), and Ar with a molar H_2/CO/Ar ratio of 6/3/1 at 543 K, 5 MPa, and a *WHSV* of 4000 cm^3 g_{cat}^{-1} h^{-1}. In order to compare different catalysts at the same CO conversion level (7−8%), the feed flow rate was varied (*WHSV* = 500−48 000 cm^3 g_{cat}^{-1} h^{-1}). The data relative to this comparison represent the average value of 4−5 measurements taken between ca. 11 and 15 h on stream. Kinetic investigations were additionally conducted over the best catalyst by varying the temperature, pressure, H_2/CO ratio, and *WHSV* between 473 and 593 K, 2 and 7 MPa, 0.2 and 4, and 500 and 32 000 cm^3 g_{cat}^{-1} h^{-1}, respectively. The results reported were determined averaging values obtained from 4−5 measurements between ca. 6−10 h on stream. The best catalyst was also assessed in a 100 h test.

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Table 1. Characterization Data of the Carbon Carriers, Reduced Supported CuFe Catalysts, and a Reduced Bulk CuFe Catalyst

catalysts	bulk CuFe content[a] (wt %)	bulk molar Cu/Fe ratio[a] (−)	surface molar Cu/Fe ratio[b] (−)	$S_{BET,reduced}$[c,d] ($m^2\ g^{-1}$)	$V_{pore,reduced}$[c,e] ($cm^3\ g^{-1}$)	$d_{Cu,XRD}$[f] (nm)	$d_{Cu,STEM}$[g] (nm)	$d_{Fe,STEM}$[g,h] (nm)
AC-2	5.2	2.05/1		866 (1154)	0.74 (1.05)	23.9 (22.4)	28.1	6.9
CNF_p-2	4.8	2.11/1		184 (201)	0.43 (0.43)	13.1 (25.0)	14.1	15.5 (15.9)
CNT-2	4.9	1.93/1		190 (168)	0.60 (0.77)	7.2 (9.7)	5.6	2.7
CNF-2	4.9	1.93/1	0.8	28 (28)	0.08 (0.08)	9.5 (9.9)	11.9	4.8
CP-2		1.94/1		6 (63)	0.02 (0.16)	37.8 (38.8)		

[a]ICP-OES, XRF for CP-2. [b]XPS. [c]Data for carriers and CP-2 in calcined form in brackets. [d]BET method. [e]*t*-plot method. [f]Determined from the (111) reflection of Cu in the XRD pattern using the Scherrer equation. Data for used samples in brackets. [g]Determined from STEM images. [h]Determined from the (011) reflection of Fe in the XRD pattern using the Scherrer equation and shown in brackets when available.

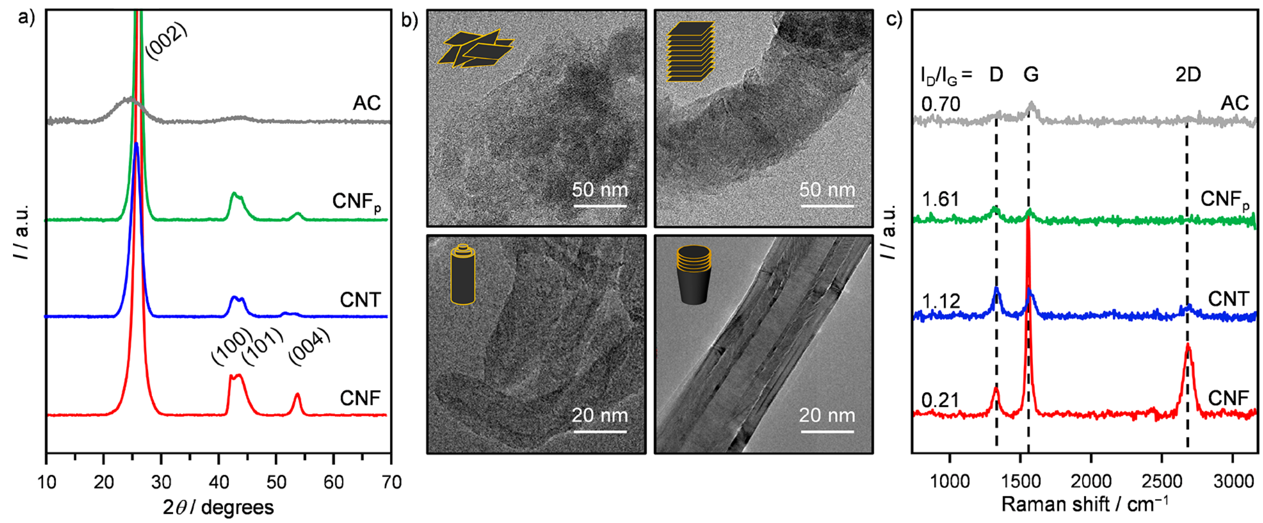

Figure 1. (a) XRD patterns, (b) TEM images, and (c) Raman spectra of AC, CNF_p, CNT, and CNF.

In this case the data refer to a single measurement. The CO conversion (X_{CO}) was calculated using eq 1

$$X_{CO} = \frac{n_{CO,in} - n_{CO,out}}{n_{CO,in}} \times 100\% \quad (1)$$

where $n_{CO,in}$ and $n_{CO,out}$ are the molar flows of CO (expressed in mmol h^{-1}) at the inlet and outlet of the reactor, respectively. The selectivity to product i (S_i) was calculated using eq 2

$$S_i = \frac{n_{i,out}N_{c,i}}{\sum n_{i,out}N_{c,i}} \times 100\% \quad (2)$$

where $n_{i,out}$ and $N_{c,i}$ are the molar flow of product i and the number of carbon atoms in product i, respectively. The selectivity to HA was obtained summing the individual selectivities to alcohols with 2 or more carbon atoms, while that to hydrocarbons summing the individual selectivities to hydrocarbon with 1 or more carbon atoms. The space time yield of HA (STY_{HA}) expressed in $g_{HA}\ g_{cat}^{-1}\ h^{-1}$ was calculated using eq 3

$$STY_{HA} = \sum S_{j,HA} MW_{j,HA} \frac{X_{CO} n_{CO,in}}{m_{cat}} \quad (3)$$

where m_{cat} is the mass of the catalyst and $MW_{j,HA}$ is the molecular weight of (higher) alcohols containing j carbon atoms. The carbon balance was determined according to eq 4 and was always higher than 95%

$$\varepsilon_C = \frac{n_{CO,in} - \sum n_{i,out}N_{c,i}}{n_{CO,in}} \times 100\% \quad (4)$$

3. RESULTS AND DISCUSSION

3.1. Effect of the Carrier Structure. The porous, structural, and morphological properties of the four carbon carriers studied were assessed by N_2 sorption (Table 1 and Figure S1 in the Supporting Information), XRD, TEM, and Raman spectroscopy. Surface areas and pore volumes were comparable to literature data for all materials.[8,32,33] The surface area was the largest for AC, which was followed by CNF_p, CNT, and finally CNF. The diffractogram (Figure 1a) of AC evidenced two very broad peaks, highlighting the amorphous nature of this sample, while the patterns of CNF, CNF_p, and CNT featured reflections specific to graphitic carbon at 2θ = 24.4°, 42.4°, 43.8°, and 54.0°, corresponding to the (002), (100), (101), and (004) planes, respectively. The diffraction peak of the (002) plane is broader and slightly shifted toward lower angles for CNT, indicating a larger d_{002} spacing. TEM images evidence typical morphologies for the carriers (Figure 1b), comprising agglomerated particles (AC), fibers with an outer diameter of ca. 80 nm made of graphene layers stacked perpendicular to the fiber axis (CNF_p), hollow fibers with inner and outer diameters of ca. 20 and 50 nm constructed from conically-shaped graphene sheets stacked in parallel to the fiber axis (CNF), and multiwalled tubes with a similar hollow-core structure to CNF but with much smaller inner (6 nm) and outer (18 nm) diameters (CNT). The Raman spectra (Figure 1c) of the four supports evidenced two characteristic bands at about 1580 and 1330 cm^{-1}. The former is attributed to the stretching of the bonds between the sp^2 C atoms in graphite (G band), while the latter is ascribed to the stretching of the bonds of the same atom at defect positions (distorted atoms at edges or in the lattice, D band).[34] The ratio of the intensity of the D and G bands (I_D/I_G), which is a measure of the density of defect sites, increased in the order CNF (0.21) < AC (0.70) < CNT (1.12) < CNF_p (1.61), in line with the literature.[35−38] The high defect concentration in CNF_p originates from the higher fraction of edges exposed by the assembly of small graphene sheets. In contrast, CNT comprises long sheets rolled up in tubes, exposing defects only at their openings. The amount of defects in CNF is even lower due to the high order in the stacking of the individual graphene layers. The ordered carbon structure of CNF and CNT is further corroborated by the presence of the overtone band at about 2700 cm^{-1} (2D band).

CuFe catalysts supported on these carbonaceous materials with nominal metals loading of 5 wt % and molar Cu/Fe ratio of 2 were prepared by a sol−gel method previously used by us to synthesize CoMo catalysts with enhanced metal dispersion and intimacy[8] and activated by reduction in diluted H_2 at 673 K. A bulk CuFe catalyst with a molar Cu/Fe ratio of 2 was prepared by coprecipitation, calcined in air at 723 K for 3 h, and activated under the same conditions to serve as a reference. On the basis of elemental analysis by ICP-OES or XRF, the actual metals loadings and ratios of the catalysts were very close to the nominal values. The surface area and pore volume of the supported catalysts were slightly lower or equivalent (CNF-2) to those of the bare carriers, in line with the low content of metals deposited (Table 1). The surface area of CP-2 was 63 $m^2\,g^{-1}$ after calcination, which is in good agreement with the literature[20] and decreased to 6 $m^2\,g^{-1}$ after the subsequent reduction step. The XRD patterns of the reduced catalysts (Figure S2) revealed the presence of metallic Cu (2θ = 43.5° and 50.6°) in all cases. On the basis of the Scherrer equation, the Cu particle size followed the trend CP-2 (37.8 nm) > AC-2 (23.9 nm) > CNF_p-2 (13.1 nm) > CNF-2 (9.5 nm) > CNT-2 (7.2 nm, Table 1). Reflections specific to metallic Fe (2θ = 44.8° and 65.2°) were identified in the diffractograms of CNF-2, CNT-2, and CNF_p-2, but only those produced by the latter sample were intense enough to enable a reliable estimation of the particle size (16.0 nm). Indeed, the most intense reflection of Fe overlaps with those of the carbon supports. However, this indicates that Fe was more finely dispersed on CNF and CNT. CP-2 comprised Fe_3O_4 (32.8 nm) as the only Fe-containing phase. Inspection of the supported catalysts by STEM-EDX (Figure 2 and Figure S6a for CNF-2) evidenced a high Fe dispersion and the presence of Cu nanoparticles having quite similar sizes to those determined by XRD. Minor discrepancies might be related to different sampling capability of the two techniques. Specifically, the average particle sizes of Cu and Fe were estimated at 5.6 and 2.7 nm for CNT-2, 11.9 and 4.8 nm for CNF-2, 14.1 and 15.5 nm for CNF_p-2, and 28.1 and 6.9 nm for AC-2 (Table 1). Since the metals particles are predominantly located in the hollow cores in the CNT- and CNF-supported catalysts, their smaller size compared to the AC- and CNF_p-supported systems might be attributed to a confinement effect. In line with this, they are moderately larger within CNF, which features wider channels. The EDX maps of AC-2, CNF-2, and CNT-2 also indicate that all Cu particles visualized are in contact with the Fe phase. In contrast, the contact among the metals is limited in CNF_p-2 with many Cu particles remaining isolated.

The performance of these catalytic systems in the direct conversion of syngas to HA was investigated at a temperature

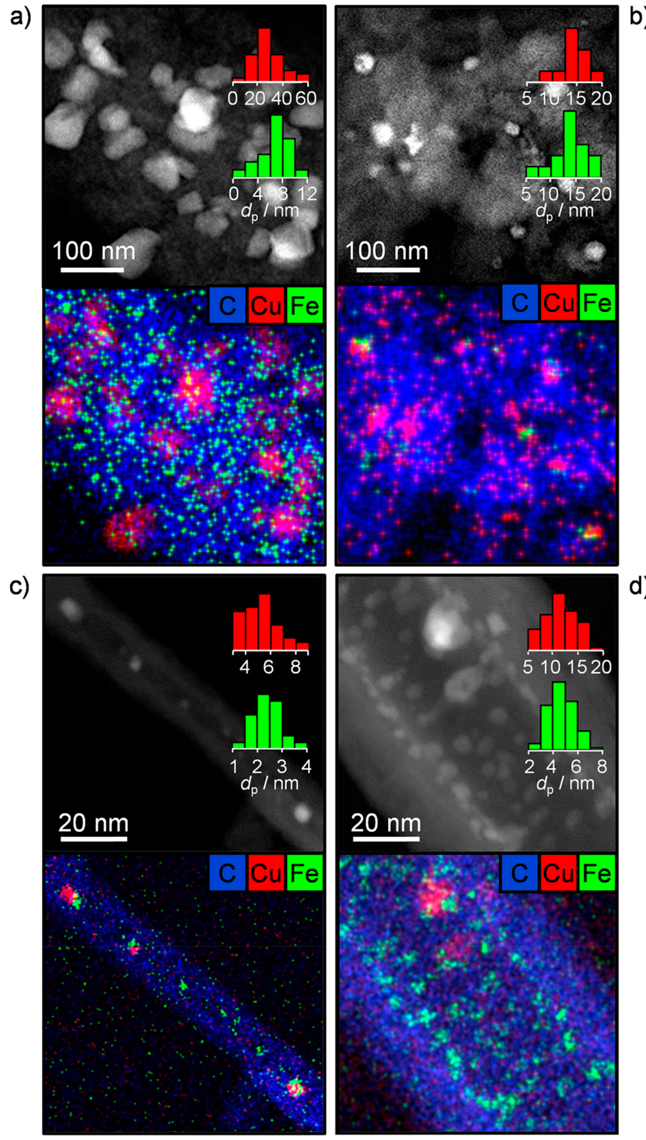

Figure 2. STEM images and EDX maps of C (blue), Cu (red), and Fe (green) of reduced (a) AC-2, (b) CNF_p-2, (c) CNT-2, and (d) CNF-2. Particle size distributions of Cu and Fe in red and green, respectively.

of 543 K, a total pressure of 5 MPa, and a molar H_2/CO ratio of 2 (Figure 3 and Table S1). The supported catalysts exhibited very different activity upon an initial testing at *WHSV* of 4000 $cm^3\,g_{cat}^{-1}\,h^{-1}$, with AC-2 and CNF_p-2 being almost an order of magnitude less active than the other two systems. Accordingly, the *WHSV* was varied between 500 and 32 000 $cm^3\,g_{cat}^{-1}\,h^{-1}$ to compare their selectivity at the same CO conversion (X_{CO} = 7−8%). A very diverse product distribution was observed. CNF-2 achieved the highest HA selectivity (31%) and was followed by CNT-2 (21%), CNF_p-2 (11%), and AC-2 (2%). The same trend was identified for the olefins (HC=) selectivity, reaching up to 18% for CNF-2 and as low as 1% over AC-2. The methanol selectivity was essentially the same for CNF-2 and CNF_p-2 (ca. 12%), while that of CNT-2 was much lower (3%). The paraffins (HC−) selectivity was the highest in CNT-2 (53%), inferior and similar for CNF-2 (32%) and CNF_p-2 (30%), and very low for AC-2 (7%). CO_2 production vastly deviated from one system to another. AC-2 and CNF_p-2 displayed an extremely high

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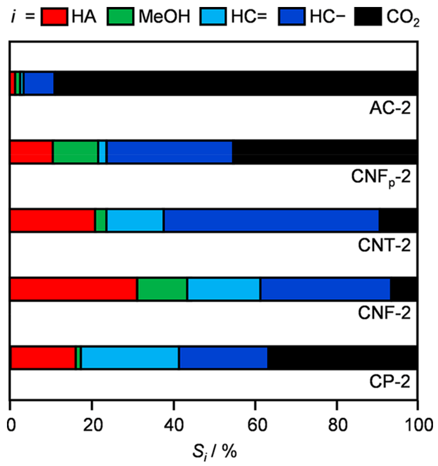

Figure 3. Selectivity to product *i* at a CO conversion level of 7−8% over CuFe catalysts in bulk form and supported on different carbonaceous materials. Reaction conditions: 543 K, 5 MPa, and molar H_2/CO = 2.

CO_2 selectivity (88% and 45%, respectively). In stark contrast, the formation of this much undesired product was very limited over CNT-2 and CNF-2 (9% and 6%, respectively). CP-2 exhibited a very high CO conversion under standard conditions (95%) but at a rather low HA selectivity (12%). The HA selectivity was slightly higher (16%) at the same activity level as for the supported systems, but the catalyst still mainly produced hydrocarbons (46% in total with about a 1:1 ratio among alkenes and alkanes) and CO_2 (36%).

On the basis of XRD analysis of the used supported samples (Table 1 and Figure S2), Cu significantly sintered only in CNF_p-2 (from 13.1 to 25.0 nm). The negligible change observed for the other materials might be explained by the large surface area allowing more distance among particles (AC-2) and by the space limitation in the channels (CNT-2 and CNF-2).[39,40] The reflections of Fe were less intense in the patterns of the used materials compared to the corresponding reduced counterparts. This is attributed to the formation of iron carbide phases upon exposure to syngas.[41] Diffraction lines due to these species could not be identified since they overlap with those of the carriers. Thus, they were either amorphous or nanocrystalline. CP-2 after reaction comprised large Cu particles (38.8 nm) and Fe solely in the form of Fe_3O_4 and Fe_5C_2.

Aiming at establishing relations between catalyst performance and properties, it was observed that the HA selectivity and CO_2 formation were inversely and directly proportional to the size of Cu particles, respectively. Since the Cu particle size was similar in the used CNT-2 and CNF-2 but the former generated less HA, less methanol, more hydrocarbons, and, among those, less olefins, other structural characteristics shall also play a role, as expected in a bifunctional system. Since van den Berg et al.[42] found that the turnover frequency of methanol synthesis from syngas over Cu/ZnO catalysts increased with the Cu particle size reaching a plateau at ca. 9 nm, the lower selectivity to this alcohol over CNT-2 could be due to the preservation of Cu particles with smaller dimensions. The more prominent formation of hydrocarbons, especially methane and aliphatic compounds, over CNT-2 seems related to the smaller size of the Fe particles in this sample. This is in line with evidence by Torres Galvis et al.,[43] who showed that the CH_4 selectivity increases with decreasing iron carbide particle size (2−7 nm) upon FTS over Fe/CNF and the opposite trend holds for olefins generation. Since Cu is less dispersed than Fe, even if all Cu particles are closely located to Fe aggregates (as seen in the EDX maps for both samples), a significant portion of Fe remains isolated. Thus, it is advantageous that the FTS metal is more agglomerated to hinder hydrocarbon production, as in CNF-2. The high CO_2 selectivity observed over CP-2 was likely linked to the large Cu particle size. Although the latter should lead to a poor HA selectivity as for AC-2 and CNF_p-2, we put forward that the increased contact between metals due to the absence of support as the reason for the comparatively pronounced HA formation.

3.2. Impact of Cu/Fe Ratio and Synthesis Method on CNF-Supported Catalysts. Having identified CNF as a more suitable carrier, insights into the effect of the metals particle size and proximity were gathered by evaluating solids supported on this material and possessing different molar Cu/Fe ratios (1, 3, and 5) or still featuring a Cu/Fe ratio of 2 but being prepared by different synthesis methods. In particular, when using the sol−gel route a calcination step was included after impregnation and prior to reduction (CNF-2-C) or the two metals were incorporated in independent steps with intermediate reduction (Fe/Cu/CNF-2). Alternatively, deposition−precipitation using Na_2CO_3 as the precipitating agent (DP-CNF-2) or a wet deposition−reduction protocol using $NaBH_4$ as the reducing species (CR-CNF-2) were employed. According to ICP-OES, CNF-3 and CNF-5 had a slightly lower Cu loading than expected, which could be attributed to the hygroscopic nature of copper nitrate, used in larger amounts in these samples.

Table 2. Characterization Data for Reduced CNF-Supported CuFe Catalysts Prepared with Different Molar Cu/Fe Ratios and by Different Synthesis Methods

catalysts	bulk CuFe content[a] (wt %)	bulk molar Cu/Fe ratio[a] (−)	surface molar Cu/Fe ratio[b] (−)	S_{BET}[c,d] (m^2 g^{-1})	V_{pore}[c,e] (cm^3 g^{-1})	d_{Cu}[c,f] (nm)
CNF-1	5.2	0.95/1	0.8	29 (22)	0.10 (0.08)	10.6 (10.6)
CNF-3	4.6	2.80/1		28 (23)	0.10 (0.07)	18.4 (10.4)
CNF-5	4.7	4.97/1	1.2	30 (23)	0.10 (0.09)	25.2 (19.2)
CNF-2-C	5.1	2.04/1		32 (20)	0.12 (0.09)	(9.5)
Fe/Cu/CNF-2	5.1	2.22/1		28 (18)	0.10 (0.08)	11.6 (11.4)
CR-CNF-2	4.6	2.21/1	0.3	31 (24)	0.09 (0.09)	9.7 (22.8)
DP-CNF-2	4.8	2.10/1		29 (20)	0.10 (0.08)	10.9 (11.1)

[a]ICP-OES. [b]XPS. [c]Data for used samples in brackets. [d]BET method. [e]*t*-plot method. [f]Determined from the (111) reflection in the XRD pattern using the Scherrer equation.

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The measured molar Cu/Fe ratio and metals loading of the catalysts attained by distinct recipes were close to the nominal values (Table 2) except for CR-CNF-2 and Fe/Cu/CNF-2. In the first case, the incomplete reduction of the metals by $NaBH_4$ in the precursor solution (particularly Fe) is put forward as the reason, while a mismatch in the Cu loading, due to unfinished precipitation, propagating to that of Fe seems to be the cause of the deviation for the second material. The surface area of all systems was similar to or slightly larger than that of the carrier (28−32 m^2 g^{-1}). XRD analysis (Table 2 and Figures S2 and S3a) revealed the presence of metallic Cu in all samples with variable metals ratio. While its particle size was essentially the same for a Cu/Fe ratio of 1 and 2, much larger particles were present in CNF-3 (18.4 nm) and CNF-5 (25.2 nm). CuO was additionally evidenced (reflections at 2θ = 35.7° and 38.8°) in CNF-3 and CNF-5, suggesting a more difficult reduction of Cu in the Cu-rich solids. Reflections specific to metallic Fe were detected in CNF-1 and CNF-3 but not in CNF-5, and only those of CNF-1 could be used for size analysis (11.6 nm). This indicates an enhanced dispersion for this metal at lower loadings, as expected. Considering the catalysts obtained through different routes (Table 2 and Figure S3b), CNF-2-C contained metal oxides (Cu_2O and Fe_2O_3) and metallic phases (Cu and Fe) featuring small particle sizes. The formation of oxidic phases difficult to reduce suggests a lower intermixing of the metals in the working catalyst since they separated well in an additional preparation step. Fe/Cu/CNF-2 comprised larger Cu particles (11.6 nm). Although a reliable estimation of the particle size was not feasible, Fe was clearly less crystalline in Fe/Cu/CNF-2 than in CNF-2. DP-CNF-2 featured metallic Cu and Fe. The size of Cu was similar to that of Fe/Cu/CNF-2 (10.9 nm), while Fe was more aggregated (9.1 nm). The diffractogram of CR-CNF-2 showed weak Cu reflections but sharp CuO signals. This compound likely formed due to the longer exposure of the sample to air during filtration and drying. No Fe phase was identified. Analysis by STEM-EDX (Figure 4a) revealed the presence of very large Cu-based particles in this catalyst, which were mostly deposited at the outer surface of the fibers and even reached a size of ca. 200 nm. On the contrary, Fe was finely dispersed and located both in the interior and at the exterior of the fibers (Figure 4a, inset). This indicates a rapid reduction and particle growth of Cu in the solution prior to contact with the carrier and diffusion within the channel, and supports the strong role of the CNF in regulating the size of this metal, which tends to agglomerate more easily than Fe.

XPS data indicated a Cu/Fe ratio of 0.8, 0.8, and 1.2 for the reduced forms of CNF-1, CNF-2, and CNF-5, respectively, and 0.3 for CR-CNF-2 (Tables 1 and 2 and Figure S4). These results strongly deviate from the bulk values for all materials except CNF-1 and agree with the relative level of aggregation of the two metals in the samples indicated by XRD and microscopy. The high intensity of the C1*s* core level emission compared to the Cu2*p* and Fe2*p* signals hints that the two metals are mainly located inside the channels of the fibers (Table S2). Gentle milling of the surface with Ar^+ ions had no significant effect on the atomic ratio between carbon and the active metals, further supporting this hypothesis. The high intensity of the metals compared to carbon in the spectrum of CR-CNF-2 corroborates the STEM evidence that most of Cu and Fe are deposited at the outer surface of the fibers in this solid.

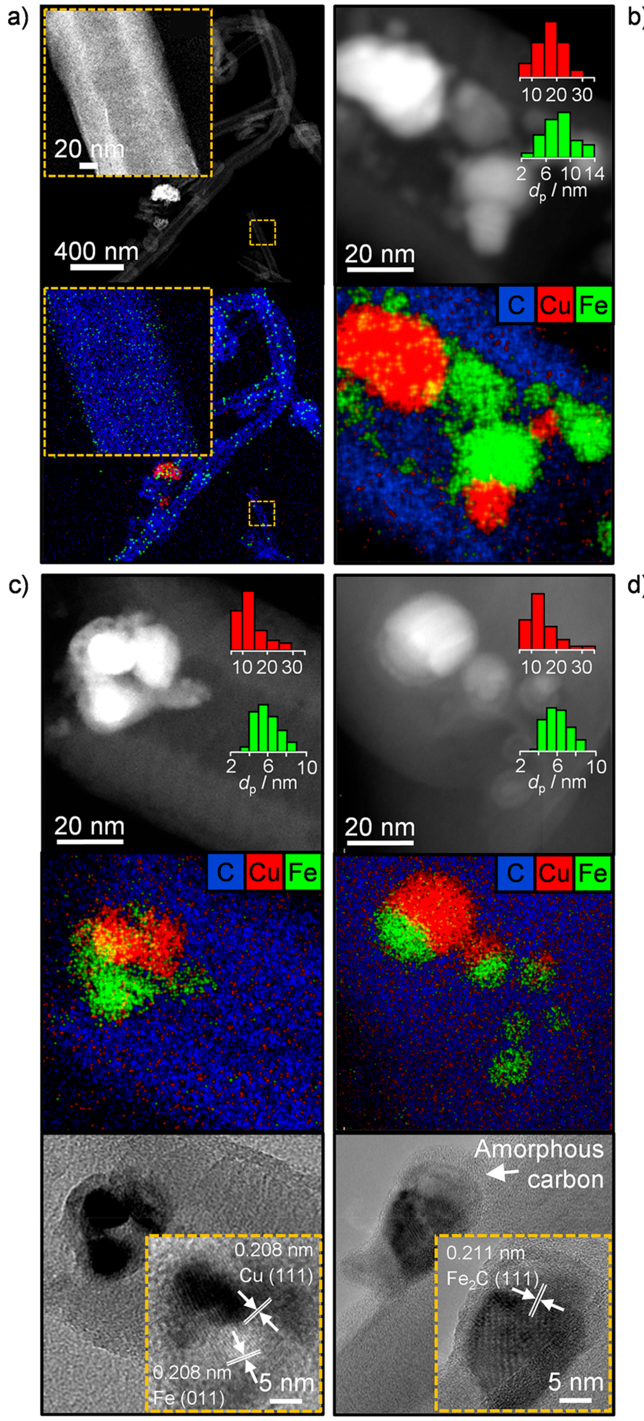

Figure 4. STEM images and EDX maps of C (blue), Cu (red), and Fe (green) of the reduced (a) CR-CNF-2, (b) K/CuFe/CNF-2-0.005, and (c) CNF-2-0.005 samples and of (d) CNF-2-0.005 after reaction. (c and d) HRTEM images are also shown. Particle size distributions of Cu and Fe in red and green, respectively.

The product distributions attained for these catalysts at the same conversion level (Figure 5 and Table S1) are quite diverse. The HA selectivity was similar for CNF-1 and CNF-3 and inferior compared to CNF-2 (21% vs 31%). In both cases, the overall fraction of hydrocarbons produced remained constant (46−48%), although less olefins were formed on the Cu-rich catalyst. The decrease in HA was accompanied by a drop in methanol production over CNF-1, which formed

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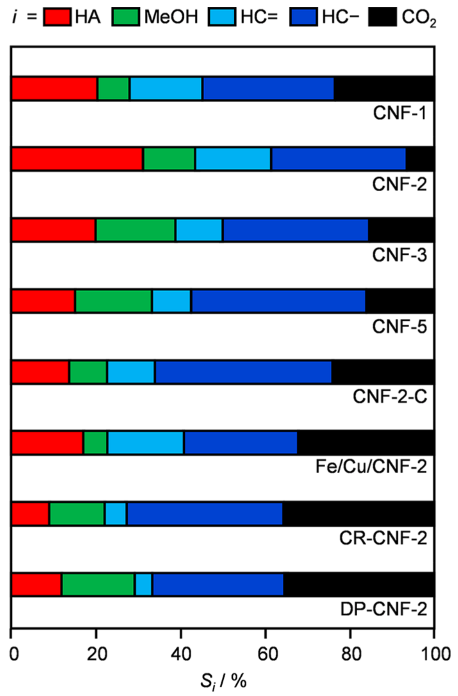

Figure 5. Selectivity to product *i* at a CO conversion level of 7−8% over CuFe catalysts supported on CNF with variable Cu/Fe ratios and prepared by different synthesis methods. Reaction conditions: 543 K, 5 MPa, and molar H_2/CO = 2.

much more CO_2. Over CNF-3, more CO_2 and methanol were generated at the expense of HA. CNF-5 formed a similar amount of methanol and CO_2 to CNF-3, but HA production was inferior due to an enhanced selectivity to hydrocarbons. As in the case of CNF-3, more paraffins were produced. This was mainly due to a boosted CH_4 formation. All materials obtained by different synthesis methods displayed a much less favorable product distribution than CNF-2. CNF-2-C and Fe/Cu/CNF-2 exhibited a higher HA selectivity than CR-CNF-2 and DP-CNF-2 (14−16% vs 9−12%). Fe/Cu/CNF-2 had a comparable performance to CNF-1 and formed less methanol and paraffins but more CO_2 than CNF-2-C. The catalysts synthesized by chemical reduction and deposition−precipitation mostly suffered from a high CO_2 and paraffins production and a reduced olefins generation. Both formed more methanol than HA. For all CNF-supported systems with Cu/Fe = 2, the trends in HA and olefins selectivites seemed to be coupled. This hints that they follow the oxygenates mechanism proposed in the FTS literature, according to which HA and olefins would derive from the same intermediate ($C_xH_yO^*$).[44]

XRD analysis of the used samples (Table 2 and Figure S3a) indicated similar properties of Cu and a moderate sintering of Fe for CNF-1 and a decrease of the particle size of Cu and no reflections for Fe in the case of CNF-3 and CNF-5. Since CuO was not detected in either of the two Cu-rich catalysts after use, the change in average dimensions for Cu is related to the formation of additional metallic particles of small size from the initial oxide phase in the reaction environment. The diffractogram of CNF-2-C evidenced the presence of Cu (9.5 nm) and of Fe_2C (Figure S3b). Again, the presence of Fe cannot be excluded, based on the overlap of its signals with those of the carrier. The XRD patterns of Fe/Cu/CNF-2 and DP-CNF-2 were nearly identical to those collected prior to reaction, clearly showing the presence of Cu (ca. 11 nm). CR-CNF-2 featured the largest Cu particles (22.8 nm) among the reference materials, and no Fe phase was detected. All samples experienced a moderate reduction in surface area after reaction, likely due to Cu sintering, carbide formation, and/or coking.

Correlating the catalytic performance to the composition, it is evident that the methanol selectivity increased with the Cu/Fe ratio up to a value of 3. CNF-5 exhibited a comparable selectivity to this alcohol to CNF-3 likely due to the much poorer Cu dispersion. In contrast to expectations, the hydrocarbon selectivity did not drop at higher Cu/Fe ratios. This is rationalized by the enhanced dispersion of Fe when present in a lower amount and higher aggregation of the more abundant Cu, which diminishes the contact between the metals. Accordingly, a larger portion of Fe acts alone, favoring methanation and FTS.[45] Since Cu and Fe in CNF-2-C feature similar dimensions to the metal phases in CNF-2, the separation between the two active metals, fostered by the intermediate calcination and formation of defined oxide compounds, is suggested to be the origin of the reduced HA formation and increased production of hydrocarbons and CO_2. The lack of intimacy among the two metals, induced by their sequential rather than simultaneous deposition, most likely rationalizes the similar behavior of Fe/Cu/CNF-2 since the dimensions of the metal aggregates are in the same range as in CNF-2-C and CNF-2. The requirement for close metal proximity for a high HA synthesis efficiency is reinforced by the poor performance of CR-CNF-2, which features the largest Cu particle and a very limited contact with Fe. This parameter is highlighted as the most important one for catalyst design. Indeed, multiple materials are characterized by a similarly small Fe and Cu size, but their HA selectivity spans though a wide range of values (Table 2). This finding justifies why very different optimal Cu/Fe ratios have been reported in relation to the best systems. The preferred value mostly fell within 1.5−3,[13,19,46] but could also go up to 10 for bulk catalysts,[18] and was as low as 0.67 for the top performer among supported materials.[25] Depending on the morphology and placement of the metal phases in the various solids, defined by the synthesis method and/or the carrier nature, a degree of proximity that permits significant HA production can be reached starting from variable relative amounts of Cu and Fe. In our case, a higher relative amount of Cu is required since this metal sinters more than Fe, but an excessive amount of the MS component should be avoided to prevent large particle formation and an undesirably high Fe dispersion.

3.3. Potassium Promotion. Potassium has been introduced into several bulk CuFe catalysts and the bimodal silica-supported CuFe system to hinder H_2 activation and thus promote C−C coupling, increasing HA formation and limiting C_1 products.[20,24] Accordingly, our study was extended to assess the impact of this promoter on the best catalyst, CNF-2. Three materials were prepared including different amounts of the alkali metal (K_2CO_3 as precursor) upon deposition of Cu and Fe so to reach a nominal K/(Cu + Fe) ratio of 0.005, 0.01, and 0.1. In addition, since the K loading is below the detection limit of many characterization techniques, we attempted to gather insights into the role of this promoter by varying its location and contact with the active metals by tailoring the synthesis method. Hence, K was incorporated before (CuFe/K/CNF-2-0.005), after (K/CuFe/CNF-2-0.005), or simultaneously to the introduction of Cu (Fe/CuK/CNF-2-0.005) or

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Table 3. Characterization Data for Reduced CNF-Supported KCuFe Catalysts with Variable K Loading and Prepared by Different Synthesis Methods

catalysts	CuFe content[a] (wt %)	bulk molar K/(Cu + Fe) ratio (−)	bulk molar Cu/Fe ratio[a] (−)	surface molar Cu/Fe ratio[b,c] (−)	S_{BET}[c,d] (m^2 g^{-1})	V_{pore}[c,e] (cm^3 g^{-1})	d_{Cu}[c,f] (nm)
CNF-2-0.1	5.0	0.07	1.98/1		24 (0.4)	0.09 (−)	20.9 (9.8)
CNF-2-0.01	4.8	0.007	2.08/1		24 (4)	0.09 (−)	11.0 (8.8)
CNF-2-0.005	4.9	0.003	2.07/1	1.2 (0.3)	24 (7)	0.09 (0.02)	11.4 (9.9)
K/CuFe/CNF-2-0.005	4.9	0.004	1.91/1	0.6	31 (28)	0.11 (0.10)	15.8 (14.7)
CuFe/K/CNF-2-0.005	5.0	0.003	1.91/1		29 (12)	0.10 (0.03)	19.3 (11.4)
Fe/CuK/CNF-2-0.005	4.4	0.004	2.02/1		28 (22)	0.12 (0.10)	
Cu/FeK/CNF-2-0.005	5.1	0.004	2.06/1		29 (15)	0.10 (0.06)	(13.5)

[a]ICP-OES. [b]XPS. [c]Data for used samples in brackets. [d]BET method. [e]*t*-plot method. [f]Determined from the (111) reflection in the XRD pattern using the Scherrer equation.

of Fe (Cu/FeK/CNF-2-0.005). The measured Cu/Fe ratio and CuFe loading of the resulting catalysts were close to the nominal values (Table 3), but the K loadings were systematically lower than expected, likely due to the presence of moisture in the K_2CO_3 precursor. On the basis of XRD (Table 3 and Figure S5a), the size of Cu crystals in CNF-2-0.005 and CNF-2-0.01 was comparable to the K-free analogue (ca. 11 nm) but twice as big in CNF-2-0.1 (20.9 nm). Similarly, the Fe phase remained well dispersed (detectable by XRD, particle size not quantifiable) in the former two solids but agglomerated in the latter (20.0 nm). It is suspected that K interacted with citric acid during the synthesis and reduced the amount of complexing agent available to Cu and Fe, thus inhibiting their dispersion, and that this produced a clearer effect only for the highest K amount. When K was introduced prior to or after the active metals, the particles of both Cu and Fe had larger dimensions than in the case of CNF-2-0.005 and CNF-2-0.01, but Cu remained more dispersed and Fe formed larger particles than in CNF-2-0.1 (Table 3 and Figure S5b). It appears that the more pronounced metal sintering results from the additional reduction step in the synthesis. By incorporating K and Cu simultaneously before Fe, both active metals were highly dispersed and the estimation of the particle size by XRD was not possible. When K and Fe were added before Cu, both transition metals could not be fully converted into their metallic forms. A mixture of metallic Cu, Cu_2O (8.6 nm), and CuO (9.2 nm) was found in the reduced sample, and Fe existed as crystalline Fe_3O_4 (8.9 nm). No K-containing phase was detected due to its very low content in all catalysts.

The reduced CNF-2-0.005 and K/CuFe/CNF-2-0.005 samples were also analyzed by STEM-EDX. In the former (Figures 4c and S6b), metal particles were mostly visible inside the CNF channels. The average Cu and Fe particle diameters were calculated to be 15.0 and 5.9 nm, respectively. Compared to the K-free CNF-2, the interaction between Cu and Fe was found to be superior, not only because of the better contact, as illustrated by the two particles detected at the pore mouth of a fiber (Figure 4c), but also because less isolated Fe particles were observed. Cu(111) and Fe(011) planes were identified based on analysis of the corresponding HRTEM image. When K was incorporated after Cu and Fe (Figure 4b), large particles of Cu (17.8 nm) and Fe (8.8 nm) formed, which featured a poorer interaction, in terms of both contact area and number of neighboring aggregates. Information about the distribution of K could not be obtained since its loading was at least 1 order of magnitude lower than the typical detection limit of EDX (ca. 0.1 wt %).

XPS (Table 3) evidenced a slightly increased surface Cu/Fe ratio (1.2) in CNF-2-0.005, but this effect was reversed when K was introduced separately (0.6 in K/CuFe/CNF-2-0.005). These deviations are in line with the more pronounced agglomeration of Fe in the first catalyst compared to the K-free counterpart CNF-2 and with the strong aggregation of Cu in the second material. Since K is added to a material that would correspond to CNF-2 in this latter case, which is characterized by a surface metals ratio of 0.8, the value of 0.6 is not surprising.

To shed further light onto the effects of K, CNF-2-0.005 and K/CuFe/CNF-2-0.005 were studied by H_2-TPD along with selected K-free samples, i.e., the counterpart of the former, CNF-2, and the poorly performing CR-CNF-2. H_2-TPD has been applied by several groups to investigate the H_2 adsorption ability of the metals and has always been conducted under ambient conditions.[13,24] In order to gather more relevant information, the adsorption and desorption of hydrogen were performed at the reaction pressure after activation of the catalyst according to the protocol applied in the catalytic tests. The same analysis procedure was applied at ambient pressure to produce reference H_2-TPD curves, which are first described here. The profiles obtained for CNF-2 and CNF-2-0.005 evidenced two weak desorption signals with maxima at ca. 690 and 870 K and a strong peak at ca. 990 K (Figure 6a). The first signal comprises a shoulder at lower temperature for the K-free sample. The amount of H_2 desorbed at these three temperatures was 0.055, 0.124, and 0.395 mmol H_2 g_{cat}^{-1} for the unpromoted material and 0.059, 0.059, and 0.172 mmol H_2 g_{cat}^{-1} for the K-containing solid. CR-CNF-2 produced a single

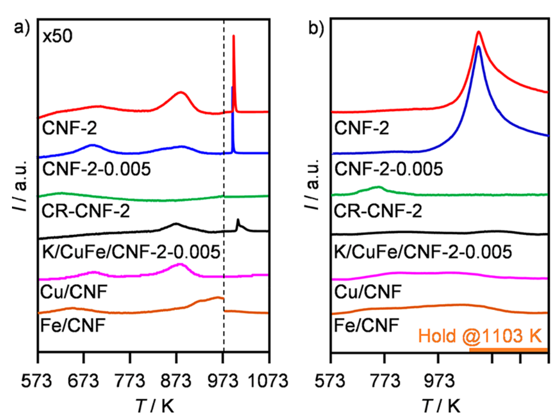

Figure 6. H_2-TPD of selected CNF-supported catalysts conducted at (a) ambient and (b) reaction pressure (5 MPa).

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and weak desorption peak located at 623 K (0.046 mmol H_2 g_{cat}^{-1}). For K/CuFe/CNF-2-0.005, two signals with maxima at 868 and 1003 K and a shoulder centered at 1065 K were identified, corresponding to 0.047, 0.116, and 0.131 mmol H_2 g_{cat}^{-1}, respectively. To assign these peaks, Cu/CNF and Fe/CNF solids with equivalent metal loadings and particle size to the materials obtained by the sol−gel method were prepared and analyzed. Fe/CNF was reduced at 773 K rather than 573 K, as the reduction of Fe does not take place at the lower temperature in the absence of Cu. On the basis of these analyses, the signals below 973 K are mainly due to H_2 desorption from Cu and that above 973 K to desorption from Fe for CNF-2, CNF-2-0.005, and K/CuFe/CNF-2-0.005. The sharpness and higher temperature associated with the Fe-related peaks might be explained by its narrow particle size distribution and interaction with Cu. The signal at 623 K in the curve of CR-CNF-2 is predominantly attributed to Fe in view of the much higher dispersion of this metal in this sample compared to Cu. It is worth noting that H_2 desorption from Cu was never observed above 430 K for previously reported materials, suggesting the existence of support effects.[47] On the basis of the analysis detailed above, Cu and especially Fe display a lower H_2 adsorption capacity in the presence of K (CNF-2 vs CNF-2-0.005), in agreement with earlier indications. The scarce H_2 adsorption ability of CR-CNF-2 is tentatively related to the very poor contact among Cu and Fe, limiting the reduction of the latter upon catalyst activation. For K/CuFe/CNF-2-0.005 the lower dispersion of both Cu and Fe is thought to be the cause of the moderate amount of H_2 evolved. When H_2-TPD was conducted at high pressure (Figure 6b), the desorption peaks shifted toward higher temperatures, as expected. Both CNF-2 and CNF-2-0.005 produced a weak peak at ca. 800 K and an intense peak centered at 1103 K (during the holding time). The profile of K/CuFe/CNF-2-0.005 shows desorption peaks at temperatures similar to the previous samples, whereas that of CR-CNF-2 features only one signal at 730 K. On the basis of the curves collected for Cu/CNF and Fe/CNF showing peaks at 824 and 995 K and 724 and 1063 K, respectively, the first peak is attributed to H_2 desorption from Cu and the second to the evolution of the probe molecule from both metals. The overall amounts of H_2 desorbed were 0.539, 0.648, 0.059, and 0.147 mmol H_2 g_{cat}^{-1} for CNF-2, CNF-2-0.005, CR-CNF-2, and K/CuFe/CNF-2-0.005, which are slightly, almost 3 times, and moderately higher and about one-half compared to the corresponding measurements at ambient pressure, respectively. The anomalous behavior of the latter catalyst is tentatively related to more pronounced Fe sintering upon activation under pressure, further limiting its contact with the rather agglomerated Cu. Since the signal tends to return to the baseline more rapidly for CNF-2-0.005 than for the K-free counterpart, even if more H_2 is adsorbed in the presence of the alkali species, its binding to the metals was weaker. In order to rationalize these results, which are in contrast with the evidence collected at low pressure, the catalyst reactivation step prior to the adsorption of H_2 was investigated (H_2-TPR, Figure S7). The curves obtained at ambient pressure evidenced a higher H_2 consumption for CNF-2 than for CNF-2-0.005. The higher reduction degree attained by the unpromoted system is in line with the higher amount of H_2 activated under atmospheric conditions. The profiles acquired at 0.5 MPa evidenced a more significant H_2 consumption, as expected, but the curve of CNF-2-0.005 displayed a visibly stronger peak for the reduction of Fe_3O_4 to Fe than in the case of CNF-2. Accordingly, it is deduced that the promoted system adsorbed more H_2 at high pressure owing to the more extensive reduction of Fe under pressure especially fostered by the greater proximity of this phase to Cu. The data also indicate that H_2 activation (by Fe) cannot be significantly promoted by pressurization for samples with intrinsic structural limitations. This demonstrates the importance of characterization under reaction conditions to provide relevant information about the catalyst. Analogous measurements using CO to study the adsorption of the second reactant were not performed, since CO-TPD curves collected at ambient pressure in earlier studies on Fe-based catalysts indicate that possible contributions from metallic, oxidic, and carbide species cannot be resolved.[48]

Testing of the promoted catalysts indicated that the simultaneous addition of small amounts of K to Cu and Fe during synthesis (CNF-2-0.005 and CNF-2-0.01) increased the selectivity toward HA, alkenes, and CO_2 and reduced that toward methanol and alkanes. The HA selectivity reached 37% over CNF-2-0.005 and matched the selectivity of the state-of-the-art CuFe catalyst (47%) over CNF-2-0.01 (Figure 7).[26]

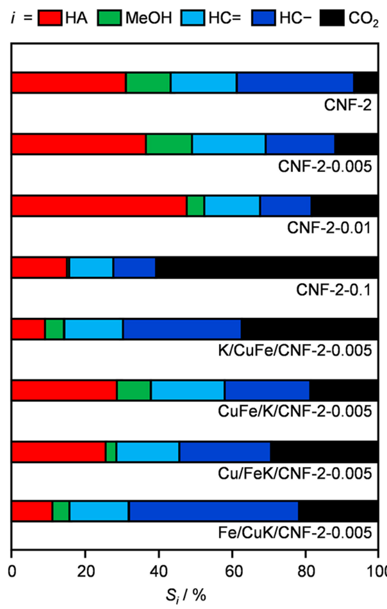

Figure 7. Selectivity to product i at a CO conversion level of 7−8% over CuFe catalysts supported on CNF and promoted with K. Reaction conditions: 543 K, 5 MPa, and molar H_2/CO = 2.

The changes in hydrocarbons selectivity led to a ca. 1:1 split among alkenes and alkanes in both cases. The boost in CO_2 formation was minimal for CNF-2-0.005 but reached a more considerable value of 18% for CNF-2-0.01. The addition of greater K amounts (CNF-2-0.1) caused a massive production of CO_2 (60%) and the lowering of the HA selectivity to 15%. The alkali effect of minimizing methanol formation was more pronounced at higher contents. The selectivity to this alcohol dropped to only 1% over the most K-rich sample. It is worth noting that CNF-2-0.005, CNF-2-0.01, and CNF-2-0.1 displayed a very different activity, attaining a CO conversion of 48%, 70%, and 12%, respectively, at a *WHSV* of 4000 cm^3 g_{cat}^{-1} h^{-1}. Hence, the highest *STY* of HA was reached over CNF-2-0.005 (0.28 g_{HA} g_{cat}^{-1} h^{-1}). When K was introduced prior to the deposition of the active metals, the HA selectivity was similar to that of the K-free CNF-2 (30%) but more CO_2

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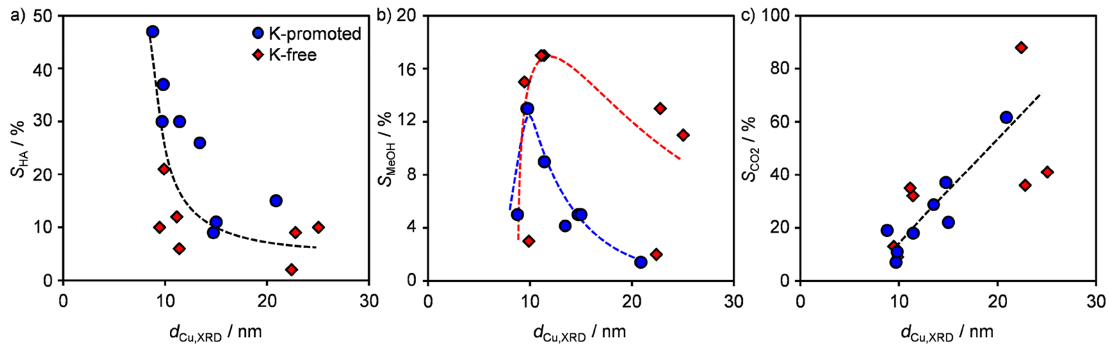

Figure 8. Selectivity to (a) HA, (b) MeOH, and (c) CO_2 as a function of crystallite size of Cu determined by XRD.

was formed (18% vs 6%) at the expense of paraffins and methanol. When the promoter was added after Cu and Fe, the HA selectivity reduced drastically to 9% and CO_2 was the most abundant product (37%). Fe/CuK/CNF-2-0.005 and Cu/FeK/CNF-2-0.005 also exhibited a low selectivity toward HA (21% and 26%, respectively). The former generated more alkanes (46%) and less CO_2 (21%), while the latter produced more CO_2 (29%) and less methanol (3%).

According to XRD analysis of the used samples with different K loadings (Table 3 and Figure S5a), the size of Cu reduced slightly in CNF-2-0.005 and CNF-2-0.01 and dramatically in CNF-2-0.1, so that all samples featured aggregates of ca. 9 nm. The detectable reflections of Fe also reduced in intensity, as a consequence of redispersion or carbide formation. Diffraction lines specific to heavy paraffins were detected at 2θ = 21.3° and 23.8° in all used samples.[49] Blocking of the fibers' channels by carbonaceous products is the most plausible cause for the dramatic drop in surface area, which was more pronounced at higher K contents (Table 3). Concerning the samples promoted by K in a separate step (Table 3 and Figure S5b), the Cu phase did not undergo strong alterations in CuFe/K/CNF-2-0.005 and Fe/CuK/CNF-2-0.005, while some disaggregation took place in K/CuFe/CNF-2-0.005. For Cu/FeK/CNF-2-0.005, all Cu oxides transformed into metallic Cu and slight sintering occurred (13.4 nm). No Fe phase was detected in any of these samples. The reflections due to paraffins were observed for CuFe/K/CNF-2-0.005 as well as for Cu/FeK/CNF-2-0.005.

STEM-EDX and HRTEM were conducted to assess the metals morphology, distribution, and speciation in CNF-2-0.005 after reaction. The mapping pointed to a higher fraction of particles located at the fibers' outer surface compared to the reduced solid (Figure S6d). The particle size and interaction among Cu and Fe were preserved (Figure 4d). A core−shell structure was detected in Fe-containing particles. On the basis of HRTEM, these comprised amorphous carbon in the outer layer and Fe_2C as the inner core. The (111) plane of this carbide was identified upon fringe analysis. To further investigate the state of iron in the used catalyst, H_2-TPR was conducted, monitoring methane formation (due to the reaction of carbon with H_2, Figure S9a) and H_2 consumption (Figure S9b). In both profiles, a broad peak centered at 789 K is evident with shoulders at lower (ca. 700 K) and higher temperatures (ca. 823 K), which were attributed to the conversion of Fe_2C, amorphous carbon, and Fe_5C_2, respectively.[50,51] An additional peak at ca. 573 K and a further shoulder at low temperatures were detected in the hydrogen consumption curve, which were ascribed to the reduction of Fe_3O_4, possibly formed by the action of water in the reaction, based on the results of the H_2-TPR analysis of the bare CNF support and of the same catalyst in reduced form. Besides for a very weak and broad methane signal centered at 573 K, produced by the latter samples and likely related to a minor amount of amorphous carbon present in the carrier, the H_2 profile of reduced CNF-2-0.005 was identical to that in Figure S7a, indicating the reduction of CuO, Fe_2O_3, and Fe_3O_4 produced upon exposure of the catalyst to air after activation. On the basis of XPS, the surface Cu/Fe ratio lowered from 1.2 to 0.3 upon reaction. This change is attributed to the moderately reduced size of Cu based on XRD (from 11.4 to 9.9 nm).

Overall, the catalytic and characterization data for promoted systems corroborate the conclusions drawn for the K-free materials with respect to the fact that higher HA selectivity is obtained if the Cu particles are around 10 nm in size (Figure 8a), Fe is not too dispersed, and a good interaction is established among the two active metals. Volcano relations for the methanol selectivity as a function of the Cu particle size are obtained for the catalyst with and without promoter with maximum close to that for HA synthesis (Figure 8b). CO_2 formation follows a quite linear trend with the dimensions of Cu crystallites (Figure 8c). The addition of K in small quantities enables the formation of a slightly more aggregated Fe phase, which has similar or even improved contact with Cu, minimizing alkanes formation and boosting HA and olefins production. Thus, K plays an important structural role. These dependences of product distribution on particle size are in good agreement with DFT calculations in MS and FTS. In MS, CO hydrogenation has been studied over both stepped Cu(211) and flat Cu(111) surfaces.[9] Since the adsorption energies of the HCO, H_2CO, and H_3CO intermediates and the energies of transition states were calculated to be considerably lower for the (211) surface, steps were put forward as more active sites than atoms in the terraces. Hence, smaller and defective particles would foster methanol production. In FTS, corner and edge sites of iron carbide were found to be strongly bonded by CO and crucial for methane formation, while terrace sites were indicated as superior for olefins generation.[43] A sharp increase in methane selectivity was experimentally observed over particles smaller than 4 nm. Transposing these findings to HA synthesis, we hypothesize that heavy alcohols can be more easily generated if small Cu particles, which adsorb more CO (possibly in the form of CHO species, i.e., hydrogen-assisted CO adsorption), are in contact with moderately sized Fe agglomerates, which expose more terrace sites and thus favor chain growth over methane formation. In

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addition, electronic effects by the K promoter are also likely. H_2-TPD suggests that more H_2 was bound to the metals, but the binding was less strong, which helps in slowing the kinetics of hydrogenation reactions to some extent. Unraveling the impact of K on H_2 as well as CO activation shall require further studies based on operando characterization and theoretical means. Overall, these elements further substantiate the minimal relevance of the surface Cu/Fe ratio as a stand-alone descriptor (Figure S8).

3.4. Kinetic and Stability Analysis. Kinetic aspects of the synthesis of HA were explored over the catalyst with the highest HA productivity and lowest CO_2 selectivity, CNF-2-0.005. Measurements were conducted over a wide range of temperatures, pressures, molar H_2/CO ratios, and *WHSV* relating the data to those acquired at T = 543 K, P = 5 MPa, H_2/CO = 2, and $WHSV$ = 16 000 $cm^3\ g_{cat}^{-1}\ h^{-1}$ as reference conditions. An increase in total pressure from 3 to 5 MPa (Figure 9a) doubled the CO conversion (6% vs 12%), while further raising it to 7 MPa did not produce any additional change. This is in line with previous evidence, but a detrimental role of a higher total pressure was also observed.[52,53] The HA selectivity increased up to 5 MPa and decreased at higher pressures. Below 5 MPa more CO_2 and olefins were produced, while above that pressure the formation of methanol and alkanes (mainly CH_4) was favored. The synthesis of HA is an equilibrium reaction in which the number of moles of the products is inferior to that of the reactants and is thus expected to be favored at high pressures. This holds also for methanol and hydrocarbons synthesis. Accordingly, methanol and paraffins formation were reported to be enhanced at higher pressure in the MS and FTS literature.[54,55] Over a HA synthesis catalyst, the intrinsic kinetic parameters of the different routes over the specific material will define the effective relevance of the distinct competitive transformations. On bulk CuFeZnMn catalysts,[56] the HA selectivity was shown to increase with the pressure, leveling off between 5 and 6 MPa. The performance at higher pressures was not probed. Hence, one cannot derive if a volcano relation would be relevant or not. Higher paraffins selectivity and lower HA formation were observed over Rh-based catalysts at higher pressures.[57] The authors claimed that the pressure had an effect on the ratio between metallic and oxidized Rh species, boosting the metallic ones and in turn favoring the dissociative adsorption of CO over its molecular adsorption and CH_x insertion.

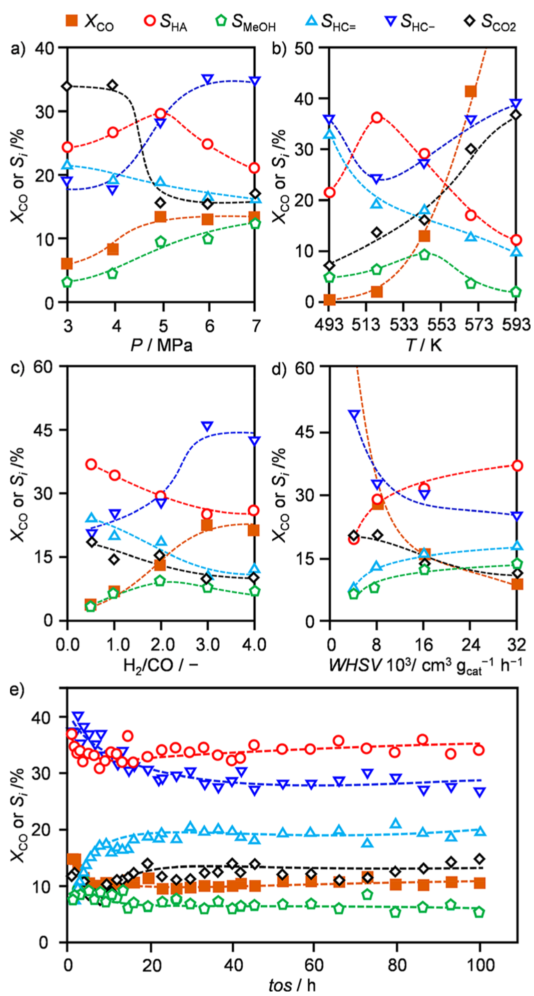

Figure 9. Selectivity to product i and CO conversion over CNF-2-0.005 as a function of (a) total pressure, (b) temperature, (c) H_2/CO ratio, (d) *WHSV*, and (e) time-on-stream (*tos*).

To study the temperature dependence (Figure 9b), its value was stepwise increased by 25 K in the range of 493−593 K. The CO conversion raised exponentially, while optimal selectivity to HA (37%) and methanol (10%) was reached at 518 and 543 K, respectively. A similar trend was reported for Rh, CuFe, and CoCu HA synthesis catalysts.[53,56,57] The formation of olefins was inhibited at high temperature, while that of CO_2 was favored due to the expected higher water−gas shift (WGS) activity. Apparent activation energies were calculated based on Arrhenius plots (Figure S10). Since it was suspected that equilibrium was attained at 593 K, the data collected at that temperature were not considered in the linearization. Values of 126 and 117 kJ mol^{-1} were determined for the synthesis of HA and methanol, respectively. Regarding hydrocarbons, methane formation was associated with the highest activation energy (142 kJ mol^{-1}), followed by alkanes (106 kJ mol^{-1}), and then alkenes (95 kJ mol^{-1}). These results are comparable to those reported for Rh and CoCu catalysts,[58,59] except for methanol. The 2-fold higher value computed in our case might be explained by the different nature of the active metals as well as by the presence of K, which was absent in the other systems.

The molar H_2/CO ratio was varied between 0.5 and 4 (Figure 9c). A higher relative H_2 concentration enhanced the CO conversion and the hydrocarbons selectivity while suppressing the selectivity toward HA and CO_2. The formation of alkanes, methane in particular, was boosted, while alkenes were produced to a lower extent. The trends observed for hydrocarbons and methanol are in line with the greater availability of H_2 and fit previous evidence in FTS and MS research.

Concerning the *WHSV*, a higher value reduced the CO conversion, as expected in view of the shorter residence time, but was beneficial to the selectivity toward HA, methanol, and olefins, the amount of which was augmented at the expense of paraffins and CO_2 (Figure 9d). The observed dependencies are consistent with previous FTS and MS data.[60] It is worth noting that the boost in HA selectivity was higher than for the other two compounds. These findings indicate that hydrogenation of carbon species and alcohol dehydration have slower kinetics, while CO insertion occurs quickly, provided that the two catalytic sites where the alkyl and CO species are formed are

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spatially close. On the basis of these observations, the differences in product distributions derived for the various supported catalysts could have been affected by the *WHSV* applied in the individual tests, which was varied in a quite large range to attain comparable activity levels. Still, the contribution by the operating conditions would mostly affect materials evaluated at less than 8000 h^{-1} (e.g., the very inefficient AC-2 catalyst), since the HA selectivity is mostly altered in this interval. For CNT- and CNF-supported systems, diffusion properties related to the presence of metal particles of distinct size, differently occluding the channels, and of coke deposits, as well as the fraction of particles in or outside the tubes will contribute to the selectivity fingerprint of the materials to some extent too. Ascertaining this aspect in a quantitative manner is very difficult and lies out of the scope of this work.

Since the stability of HA synthesis catalysts has infrequently been evaluated, CNF-2-0.005 was assessed in a 100-h catalytic run (Figure 9e). Exploiting the knowledge gained from the kinetic study, the H_2/CO ratio was reduced from 2 to 1.5 and the *WHSV* was increased from 16 000 to 32 000 $cm^3\ g_{cat}^{-1}\ h^{-1}$ to boost the HA selectivity (39%), keeping the pressure at the optimal value of 5 MPa. To compensate for the concomitant drop in CO conversion (down to 6%) under these conditions, the temperature was slightly increased (from 543 to 548 K). This enabled the attainment of a similar HA selectivity (37%) at a CO conversion of 14%. In the first 10 h of the test, the formation of alkanes and alkenes appreciably decreased and increased, respectively, and the CO conversion declined to 11% (Table S1). Thereafter, the catalytic properties remained practically unperturbed until the end of the test. The initial equilibration phase is likely associated with the formation of iron carbides. An STY_{HA} of 0.53 $g_{HA}\ g_{cat}^{-1}\ h^{-1}$ was calculated for the stable period, which is double compared to the benchmark CuFe catalyst supported on bimodal silica and promoted by K (0.23 $g_{HA}\ g_{cat}^{1}\ h^{-1}$, under optimized conditions).[25] The methanol-to-HA ratio was 0.19, which is much inferior to that reported for the most efficient catalyst reported in the literature based on bimodal silica (0.38).[25] In addition, this result was achieved maintaining the CO_2 selectivity at levels matching the lowest attained with the CuFe catalysts family under similar reaction conditions (ca. 10%),[14,46] while the benchmark system displays a CO_2 selectivity as high as 28%.[25]

CONCLUSIONS

In this study, we uncovered K-promoted CuFe nanoparticles deposited on carbon nanofibers as a bifunctional catalyst with enhanced selectivity and stability for the direct conversion of synthesis gas into higher alcohols. The catalyst design encompassed the tuning of the metals particle size and proximity through the choice of the carrier nature, of the composition and synthesis method, and of the K amount. Comparing amorphous and nanostructured commercial materials, we demonstrated that the confinement effect offered by the channels in carbon nanofibers helps to tailor the metal particle sizes, enhancing the formation of the desired products by suppressing that of C_1 compounds and CO_2. The sol−gel synthesis method and a molar Cu/Fe ratio of 2 were shown to permit the generation of Cu particles of about 10 nm in size surrounded by a smaller sized Fe phase, lowering the fraction of iron remaining isolated and thus the Fischer−Tropsch activity. The addition of tiny amounts of potassium enabled a somewhat higher degree of agglomeration of the Fe phase, enhancing its proximity to Cu. Accordingly, the selectivity toward higher alcohols was raised up to 47%. Remarkably, the formation of the most undesired product, CO_2, was only moderate, and a high fraction of olefins was obtained, i.e., ca. 50% of the hydrocarbons. In-depth analysis of the kinetic aspects of the reaction indicated lower apparent activation energies for the synthesis of higher alcohols and olefins than for methane and paraffins, in line with the best product distribution observed. The most productive catalyst achieved a selectivity to higher alcohols of 37% at a CO conversion of 14% under optimized conditions of temperature, pressure, feed H_2/CO ratio, and contact time. After an initial minor activity loss, it retained a space−time−yield of 0.53 $g_{HA}\ g_{cat}^{-1}\ h^{-1}$ during a 100-h run. Such productivity is twice as high as the state-of-the-art catalyst based on the same metals in spite of the ca. 7 times lower metals content under individually optimized conditions. Our study not only identifies a CuFe-based material with superior properties for the manufacture of relevant chemicals and fuel additives but also provides insights into key structural parameters of the catalyst, which are of general relevance in the development of complex solid systems for improved or novel applications.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acscatal.8b02714.

Additional catalytic and characterization data (PDF)

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Notes

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Supporting Information

Role of Carbonaceous Supports and Potassium Promoter on Higher Alcohols Synthesis over Copper-Iron Catalysts

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Table S1. Performance data for the catalysts studied.[a]

Catalyst[b]	$WHSV$ ($cm^3\ g_{cat}^{-1}\ h^{-1}$)	$STY_{HA[ROH]}$[b] ($g_{HA}\ g_{cat}^{-1}\ h^{-1}$)	X_{CO} (%)	S_{MeOH} (%)	S_{HA} (%)	$S_{HC=}$ (%)	S_{HC-} (%)	S_{CO2} (%)
CNF-2	4000	0.13 [0.19]	48	7	20	7	49	17
	28000	0.23 [0.37]	8	13	31	18	32	6
CNF_p-2	4000	0.01 [0.03]	8	11	11	2	31	45
	12000	0.05 [0.08]	6	9	17	8	30	36
CNT-2	4000	0.07 [0.10]	35	5	15	2	62	16
	12000	0.06 [0.08]	7	3	21	14	53	9
AC	4000	n.r.	3	1	0	1	3	95
	500	n.r.	8	2	2	1	7	88
CP-2	4000	0.15 [0.16]	95	0.5	12	4	53	31
	48000	0.19 [0.21]	7	2	16	24	22	36
CNF-1	4000	0.06 [0.10]	49	4	9	8	54	25
	24000	0.15 [0.27]	8	8	21	17	31	23
CNF-3	4000	0.07 [0.13]	34	9	15	7	57	13
	24000	0.14 [0.32]	8	19	20	11	35	14
CNF-5	4000	0.08 [0.16]	46	9	13	4	64	10
	32000	0.12 [0.32]	8	18	16	9	41	16
CNF-2-C	4000	0.05 [0.09]	38	5	10	10	59	16
	16000	0.07 [0.13]	8	9	14	11	42	24
Fe/Cu/CNF-2	4000	0.08 [0.12]	29	5	12	12	50	21
	17000	0.05 [0.07]	8	7	16	18	27	32
CR-CNF-2	4000	0.02 [0.06]	22	9	8	3	46	34
	10000	0.02 [0.07]	8	13	9	5	38	35
DP-CNF-2	4000	0.04 [0.07]	23	8	12	5	63	12
	16000	0.05 [0.16]	8	17	12	4	32	35
CNF-2-0.1	4000	0.02 [0.02]	12	0.2	13	15	12	60
	8000	0.03 [0.04]	8	1	16	12	10	61
CNF-2-0.01	4000	0.09 [0.11]	44	3	15	12	17	53
	16000	0.22 [0.25]	8	6	47	15	14	18
CNF-2-0.005	4000	0.18 [0.26]	70	6	19	7	48	20
	32000	0.28 [0.42]	8	13	37	20	19	11
	32000[f]	0.53 [0.67]	11	7	37	19	27	10
CuFe/K/CNF-2-0.005	4000	0.15 [0.21]	61	5	16	12	31	36
	20000	0.21 [0.28]	8	9	30	20	23	18
Cu/FeK/CNF-2-0.005	4000	0.08 [0.09]	37	2	14	12	17	55
	14000	0.11 [0.12]	8	3	26	17	25	29
Fe/CuK/CNF-2-0.005	4000	0.01 [0.02]	8	5	11	16	46	21
K/CuFe/CNF-2-0.005	4000	0.01 [0.02]	8	5	9	16	33	37
KCuFe/SiO_2-BM[c]	6000	0.23 [0.32]	53	23	38	11		28
CuFeMg[d]	2000	0.19 [0.28]	57	16	33	39		12
Cu_2Fe-3DOM[e]	2000	0.15 [0.19]	58	5	26	57		12

[a]Reaction conditions: 543 K, 5 MPa, and molar H_2/CO = 2. HA = higher alcohols, MeOH = methanol, HC= = olefins, and HC− = paraffins. All reported data are average values obtained between *ca.* 11−15 h on stream and the carbon balance is always higher than 95%. [b]CR = $NaBH_4$ reduction, DP = deposition precipitation, CP-2 or CP-5 = coprecipitation with Cu/Fe ratio of 2 or 5, SiO_2-BM = bimodal silica, n.r. = not relevant because negligible. [c]Reference catalyst, data retrieved from *Fuel Process. Technol.* **2017**, *159*, 436–441, reaction conducted at 593 K, 5 MPa, and molar H_2/CO = 2. STY_{HA} expressed in terms of $g_{HA}\ cm^3{}_{cat}^{-1}\ h^{-1}$. [d]Reference catalyst, data retrieved from *Catal. Sci. Technol.* **2013**, *3*, 1324–1332, reaction conducted at 573 K, 4 MPa, and molar H_2/CO = 2. STY_{HA} expressed in terms of $g_{HA}\ cm^3{}_{cat}^{-1}\ h^{-1}$. [e]Reference catalyst, data retrieved from *ACS Catal.* **2017**, *7*, 5500–5512, reaction conducted at 533 K, 4.8 MPa, and molar H_2/CO = 1. STY_{HA} determined based on the given HA formation rate and expressed in terms of mmol $g_{cat}^{-1}\ h^{-1}$. [f]Reaction conditions for the long-term test: 548 K, 5 MPa, molar H_2/CO = 1.5, and $WHSV$ = 32000 $cm^3\ g_{cat}^{-1}\ h^{-1}$. The reported data are average values obtained between 50−100 h on stream.

Table S2. Atomic concentrations of C, O, Fe, and Cu and surface Cu/Fe ratio for selected (K-promoted) CuFe catalysts supported on CNF in reduced and used forms.

Catalysts	C (at.%)	O (at.%)	Fe (at.%)	Cu (at.%)	Molar Cu/Fe ratio (-)
CNF-1	97.4	1.3	0.7	0.6	0.81
CNF-2	96.7	1.5	1.0	0.8	0.78
CNF-5	98.3	0.8	0.4	0.5	1.24
CR-CNF-2	81.8	9.4	6.8	2.0	0.29
CNF-2-0.005	96.4	1.9	0.8	0.9	1.22
CNF-2-0.005-used	98.5	1.0	0.4	0.1	0.31
K/CuFe/CNF-2-0.005	97.0	2.2	0.5	0.3	0.57

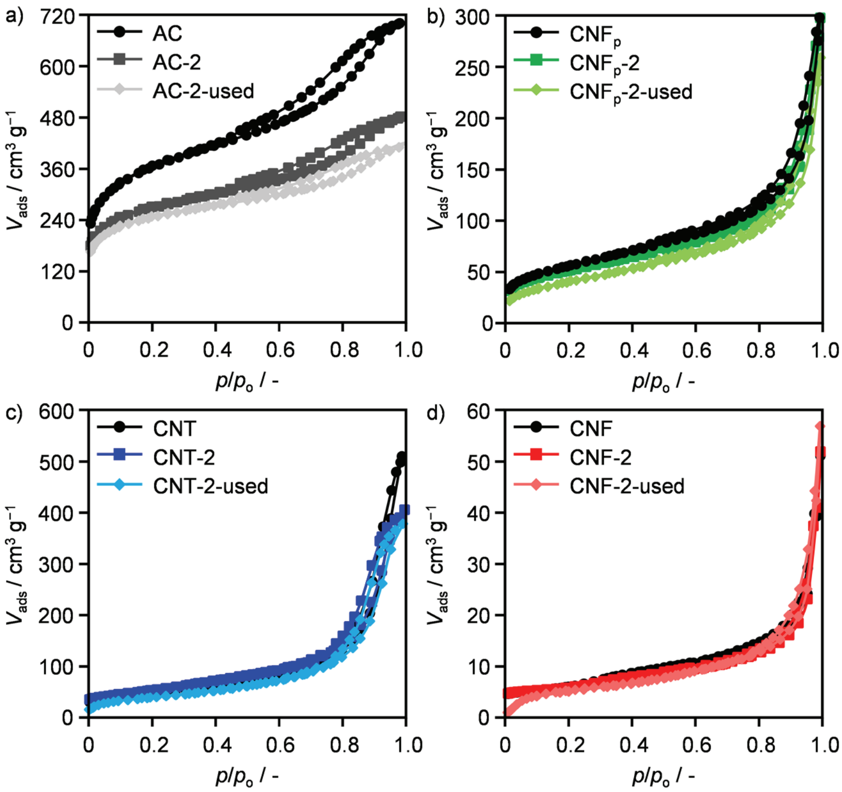

Figure S1. N_2 sorption isotherms at 77 K of a) AC, b) CNF_p, c) CNT, and d) CNF, and of the CuFe catalysts supported on these carriers in reduced and used forms.

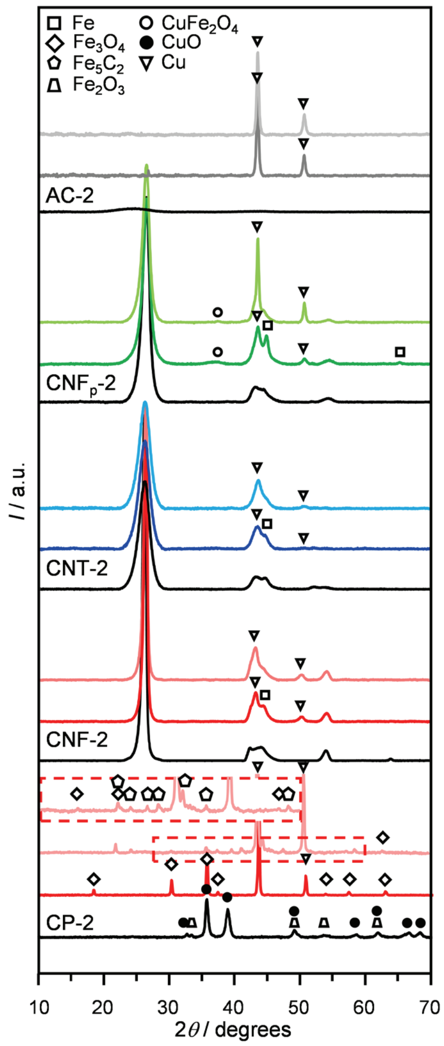

Figure S2. XRD patterns of bulk and carbon-supported CuFe catalysts in reduced (plain color) and used (pale color) forms. The patterns of the corresponding carriers and of CP-2 in calcined form are shown in black.

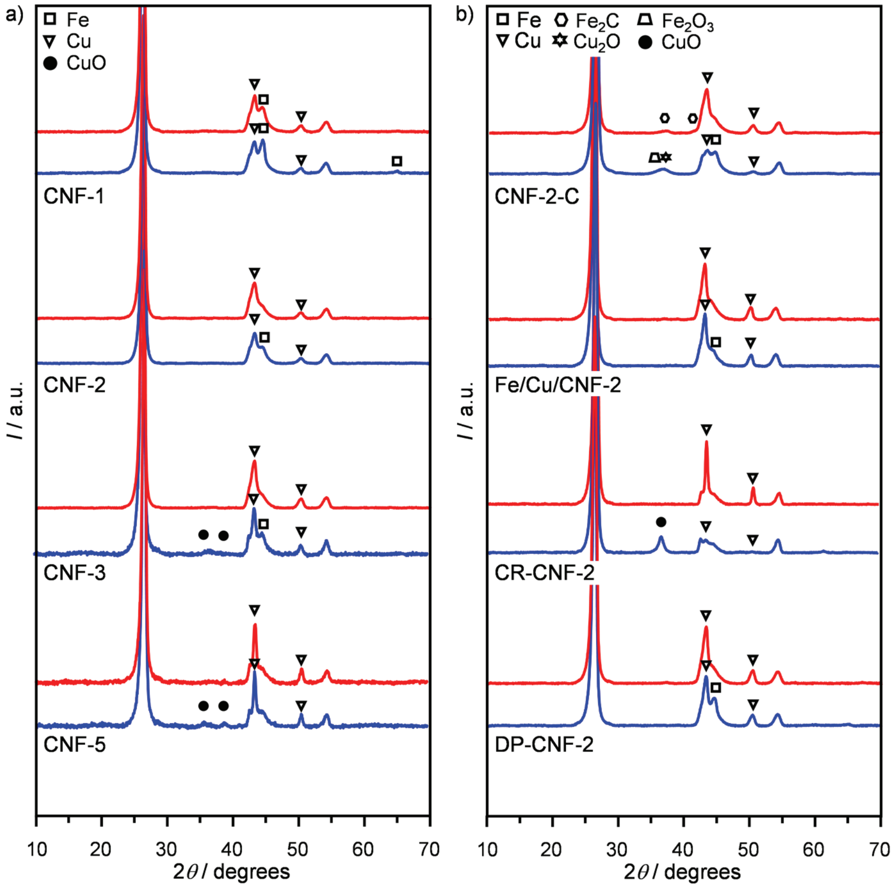

Figure S3. XRD patterns of CuFe catalysts supported on CNF and a) featuring different Cu/Fe ratios and b) prepared using different synthesis methods in reduced (blue), and used (red) forms.

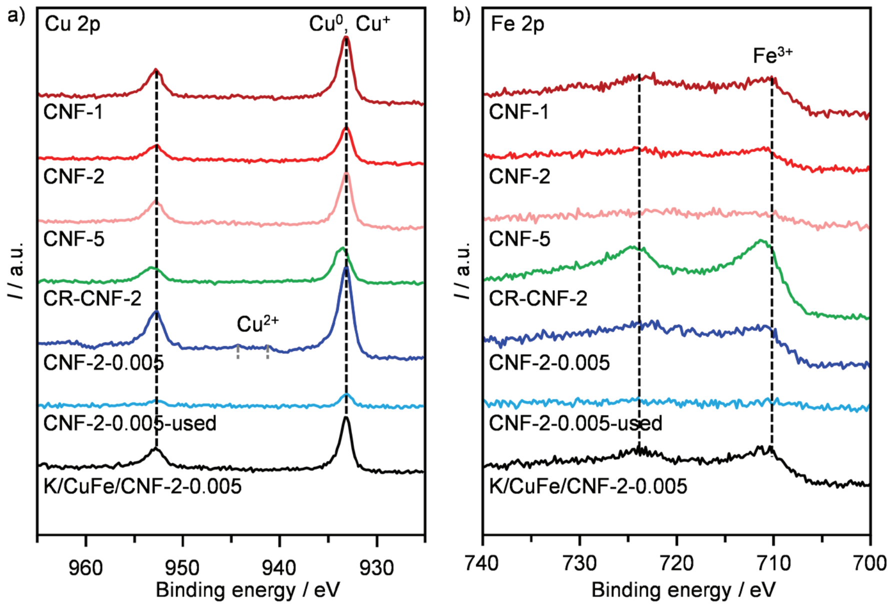

Figure S4. a) Cu2*p* and b) Fe2*p* core-level spectra of CuFe catalysts supported on CNF and featuring different Cu/Fe ratios or K as a promoter in reduced and used form.

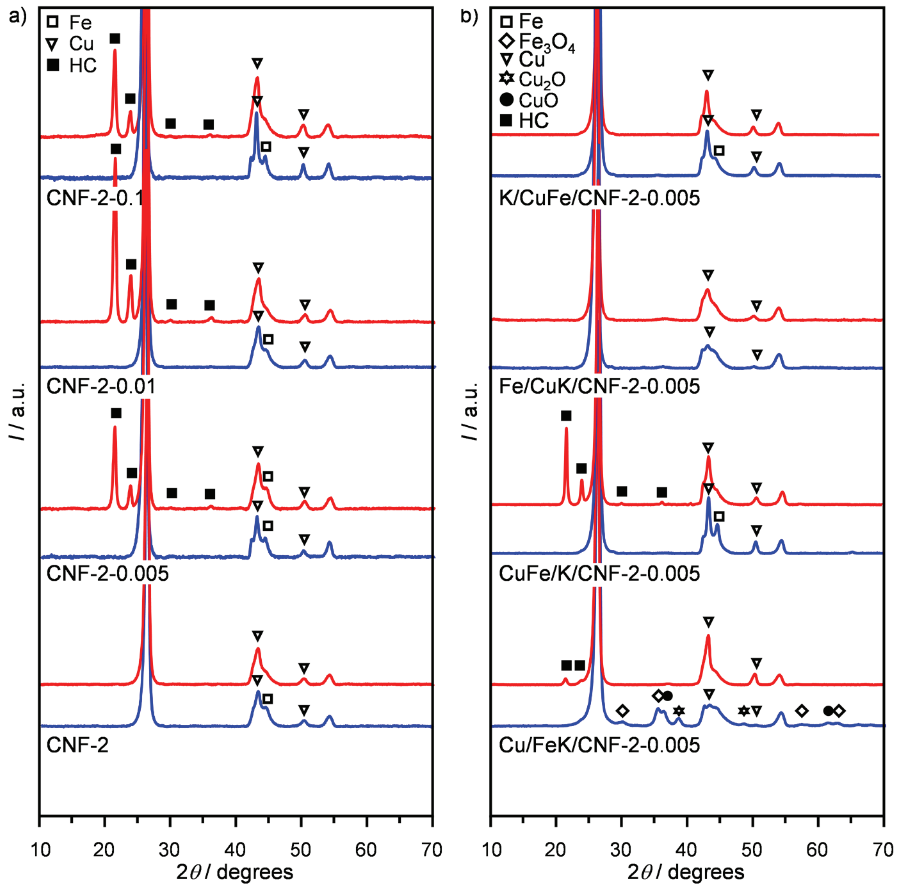

Figure S5. XRD patterns of K-promoted CuFe catalysts supported on CNF with a) distinct K contents, and b) added at different stages in the synthesis in reduced (blue) and used (red) forms.

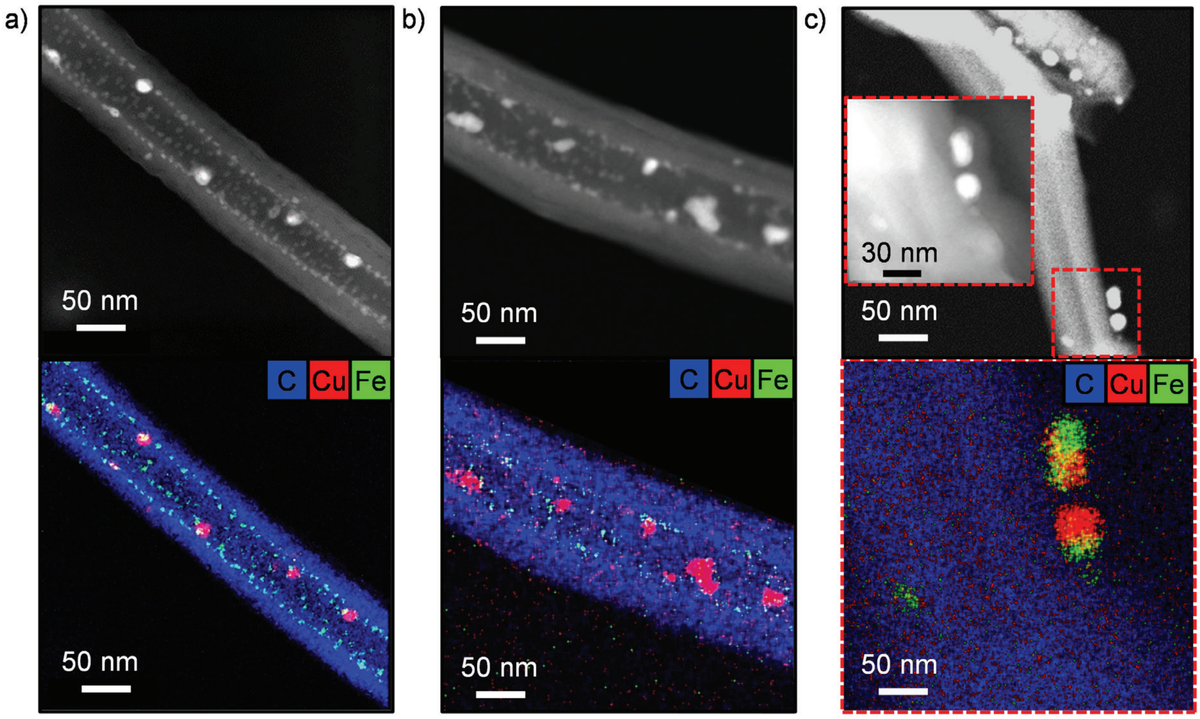

Figure S6. Additional STEM images and EDX maps of C (blue), Cu (red), and Fe (green) of a) CNF-2 and b) CNF-2-0.005 in reduced form as well as c) in used form.

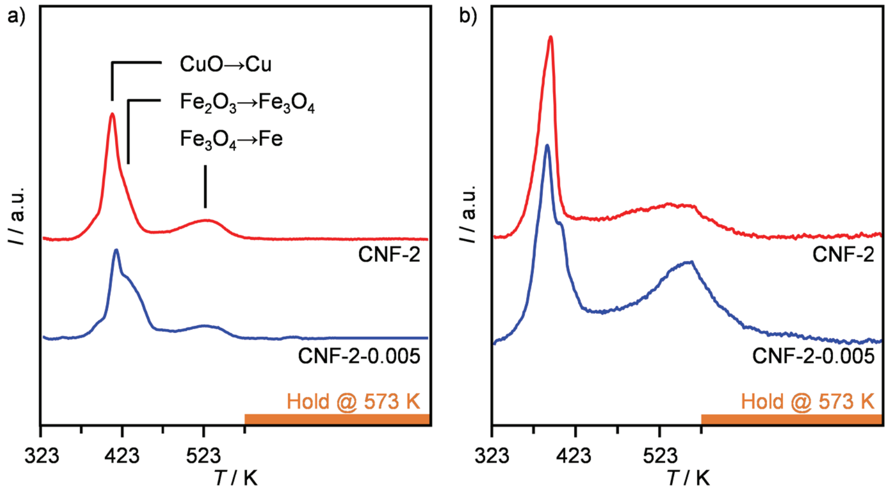

Figure S7. H_2-TPR profiles for CNF-2 and CNF-2-0.005 in reduced form measured at a) ambient pressure and b) 0.5 MPa prior to H_2-TPD analyses.

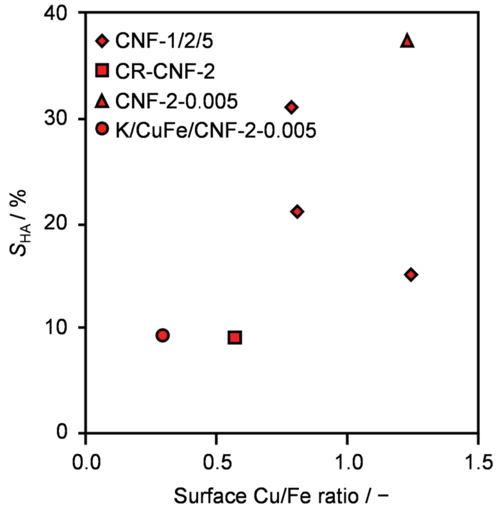

Figure S8. Selectivity to HA as a function of surface Cu/Fe ratio determined by XPS.

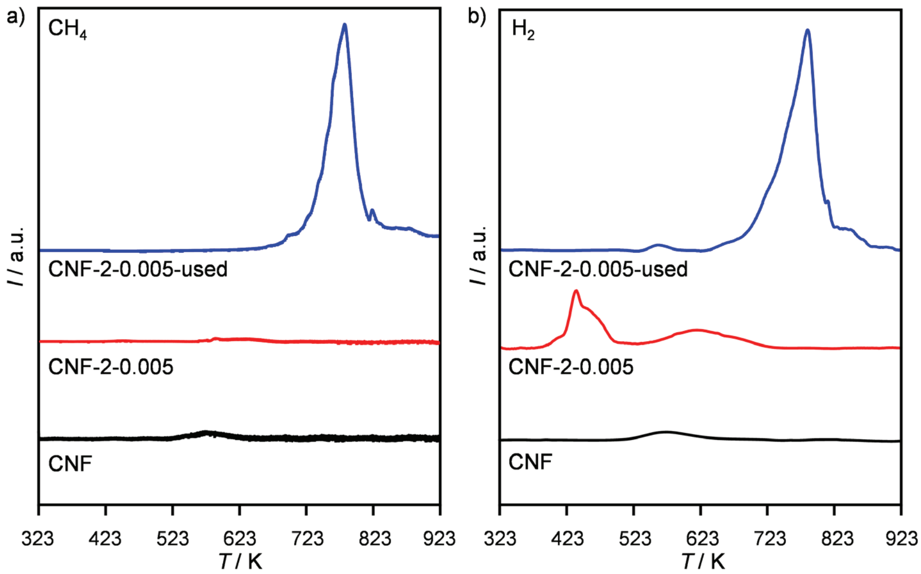

Figure S9. a) Methane formation and b) H_2 consumption by mass spectrometry (m/z = 15 and 2, respectively) upon the H_2-TPR analysis of CNF-2-0.005 in reduced and used forms and CNF.

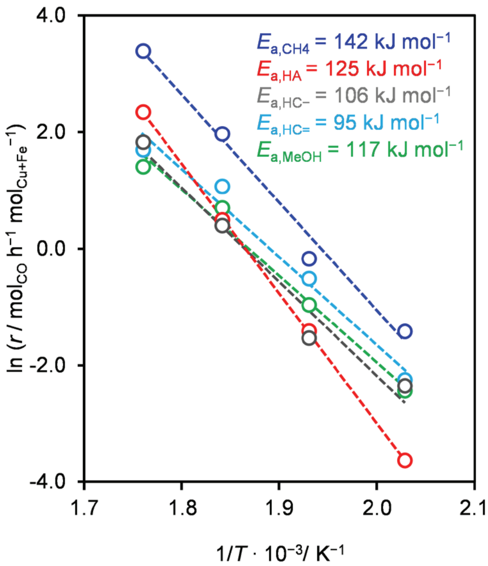

Figure S10. Arrhenius plot for CO hydrogenation to HA, methanol, methane, paraffins (excluding CH_4), and olefins over CNF-2-0.005.